

ArduPilot PNT (Positioning, Navigation, Timing) Reference Audit

Scope. This document is an audit-grade, read-only static-analysis catalog of every location in the ArduPilot firmware codebase (repository root containing ArduCopter/, ArduPlane/, Rover/, ArduSub/, AntennaTracker/, Blimp/, libraries/, modules/) where Positioning, Navigation, and Timing (PNT) data is handled, together with a two-layer dependency map for every catalogued reference. GROUP 1 catalogs core references — code that directly implements, reads, writes, or processes PNT data. GROUP 2 catalogs indirect / relational references — code that does not directly read or write PNT data but whose behavior changes as a result of PNT state. All file paths are repository-root-relative; every code snippet is reproduced VERBATIM from the cited path and line range so each row is independently verifiable. No source file is modified by this audit.

Legend

Role (Group 1) Source = produces/writes PNT state from a measurement | Transform = fuses/derives/converts PNT state (EKF fusion, GPS blending, geodetic↔NED) | Sink = only reads PNT state to drive control/behavior.

Group 2 columns Trigger Condition (what invokes the code) | PNT State Observed (the exact accessor/value read, e.g. gps.status(), ahrs.have_inertial_nav(), EKF variance) — the discriminator from Group 1 | Behavior Change Description (how behavior changes).

Dependency Type (Layer 1) Data dependency (shared data value/struct field) | Function call (direct call) | Inheritance / Interface (backend overrides a virtual frontend interface) | Shared global / singleton (access via an AP:: accessor — AP::gps(), AP::ahrs(), AP::scheduler(), AP::rtc()).

Notes flags [CIRCULAR] circular dependency (e.g. AHRSS↔EKF) | [SHARED-STRUCT] PNT fields packed with non-PNT fields (extraction risk) | [DUPLICATED] copy-pasted PNT logic across locations | [MULTI-CATEGORY] spans >1 PNT category (emitted once per table) | [ABSENT] expected PNT element not found where checked.

Cross-reference Each main table owns a monotonic, per-table # space (1,2,3...). Its dependency sub-table (the matching Xa table) reuses those exact numbers as Ref #. Each Xa table contains a Layer 1 block (direct edges) followed by a Layer 2 block (full transitive chain + final consumer + Chain Depth = the number of → hops explicitly listed in that chain).

Footprint surveyed: libraries/ = 151 subsystems; libraries/AP_GPS/*.{cpp,h} = 44 files (14 receiver backends each catalogued individually + AP_GPS_Blended + GPS_Backend base); EKF families AP_NavEKF/AP_NavEKF2/AP_NavEKF3 with AP_NavEKF3 split across 13 fusion modules (all catalogued); flight/drive-mode files = 81 (ArduCopter 29 + ArduPlane 26 + Rover 14 + ArduSub 12), enumerated in the Coverage & Explicit-Absence Register. PNT state is produced under libraries/ and consumed/acted-upon in vehicle glue code (failsafe, ekf_check, fence, flight-mode gating).

Audit-discipline flags (occurrences across the catalog): [CIRCULAR] ×8 | [SHARED-STRUCT] ×19 | [DUPLICATED] ×36 | [MULTI-CATEGORY] ×7 | [ABSENT] ×30

GROUP 1 — CORE PNT REFERENCES

Code that directly implements, reads, writes, or processes PNT data. Role is one of Source / Transform / Sink.

Table 1 — Core Positioning (GPS receiver backends, Location, inertial position, position controllers, non-GPS sources)

#	File Path	Line(s)	Function/Class	Role	Code Snippet (5-10 lines)	Notes
1	libraries/AP_GPS/AP_GPS_UBL OX.cpp	1510-1518	AP_GPS_UBLOX::parse_gp s() (NAV-POSLLH)	Source	<pre>_last_pos_time = _buffer.posllh.itow; state.location.lng = _buffer.posllh.longitude; state.location.lat = _buffer.posllh.latitude; state.have_undulation = true; state.undulation = (_buffer.posllh.altitude_msl - _buffer.posllh.altitude_ellipsoid) * 0.001; set_alt_amsl_cm(state, _buffer.posllh.altitude_msl / 10); state.status = next_fix; _new_position = true;</pre>	Receiver backend writes GPS_State.location/status from the u-blox UBX binary NAV-POSLLH. One of 14 receiver backends, each catalogued individually. [SHARED-STRUCT] target GPS_State.
2	libraries/AP_GPS/AP_GPS_NM EA.cpp	333-340	AP_GPS_NMEA::term_com plete()	Source	<pre>state.location.lat = _new_latitude; state.location.lng = _new_longitude; } state.num_sats = _new_satellite_count; state.hdop = _new_hdop; switch(_new_quality_indicator) { case 0: // Fix not available or invalid state.status = AP_GPS::NO_FIX;</pre>	Receiver backend writes GPS_State.location/status from the NMEA-0183 ASCII (GGA/RMC) sentence. One of 14 receiver backends, each catalogued individually. [SHARED-STRUCT] target GPS_State.
3	libraries/AP_GPS/AP_GPS_SBF .cpp	484-491	AP_GPS_SBF::process_mess age()	Source	<pre>state.location.lat = (int32_t)(temp.Latitude * RAD_TO_DEG_DOUBLE * (double)1e7); state.location.lng = (int32_t)(temp.Longitude * RAD_TO_DEG_DOUBLE * (double)1e7); state.have_undulation = !is_DNU(temp.Undulation); double height = temp.Height; // in metres if (state.have_undulation) { height -= temp.Undulation; state.undulation = -temp.Undulation; }</pre>	Receiver backend writes GPS_State.location/status from the Septentrio SBF binary PVTGeodetic. One of 14 receiver backends, each catalogued individually. [SHARED-STRUCT] target GPS_State.

#	File Path	Line(s)	Function/Class	Role	Code Snippet (5-10 lines)	Notes
4	libraries/AP_GPS/AP_GPS_NOVA.cpp	209-216	AP_GPS_NOVA::process_message()	Source	<pre> state.last_gps_time_ms = AP_HAL::millis(); state.location.lat = (int32_t) (bestposu.lat * (double)1e7); state.location.lng = (int32_t) (bestposu.lng * (double)1e7); state.have_undulation = true; state.undulation = -bestposu.undulation; set_alt_amsl_cm(state, bestposu.hgt * 100); </pre>	Receiver backend writes GPS_State.location/status from the NovAtel OEM BESTPOS. One of 14 receiver backends, each catalogued individually. [SHARED-STRUCT] target GPS_State.
5	libraries/AP_GPS/AP_GPS_SBP.cpp	276-283	AP_GPS_SBP::attempt_state_update()	Source	<pre> state.location.lat = (int32_t) (pos_llh->lat * (double)1e7); state.location.lng = (int32_t) (pos_llh->lon * (double)1e7); state.location.alt = (int32_t) (pos_llh->height * 100); state.num_sats = pos_llh->n_sats; if (pos_llh->flags == 0) { state.status = AP_GPS::GPS_OK_FIX_3D; } </pre>	ArduPilot_PNT_Reference_Audit.pdf Receiver backend writes GPS_State.location/status from the Swift Navigation SBP v1 POS_LLH. One of 14 receiver backends, each catalogued individually. [SHARED-STRUCT] target GPS_State.
6	libraries/AP_GPS/AP_GPS_SBP2.cpp	328-335	AP_GPS_SBP2::attempt_state_update()	Source	<pre> // state.location.lat = (int32_t) (last_pos_llh.lat * (double)1e7); state.location.lng = (int32_t) (last_pos_llh.lon * (double)1e7); state.location.alt = (int32_t) (last_pos_llh.height * 100); state.num_sats = last_pos_llh.n_sats; switch (last_pos_llh.flags.fix_mode) { case 1: </pre>	Receiver backend writes GPS_State.location/status from the Swift Navigation SBP v2 POS_LLH. One of 14 receiver backends, each catalogued individually. [SHARED-STRUCT] target GPS_State.
7	libraries/AP_GPS/AP_GPS_SIRF.cpp	178-185	AP_GPS_SIRF::parse_gps()	Source	<pre> state.status = AP_GPS::NO_FIX; } else if ((_buffer.nav.fix_type & FIX_MASK) == FIX_3D) { state.status = AP_GPS::GPS_OK_FIX_3D; } else { state.status = AP_GPS::GPS_OK_FIX_2D; } state.location.lat = int32_t(be32toh(_buffer.nav.latitude)); state.location.lng = int32_t(be32toh(_buffer.nav.longitude)); </pre>	Receiver backend writes GPS_State.location/status from the SiRF binary geodetic navigation. One of 14 receiver backends, each catalogued individually. [SHARED-STRUCT] target GPS_State.
8	libraries/AP_GPS/AP_GPS_ERB.cpp	150-157	AP_GPS_ERB::parse_gps()	Source	<pre> _last_pos_time = _buffer.pos.time; state.location.lng = (int32_t) (_buffer.pos.longitude * (double)1e7); state.location.lat = (int32_t) (_buffer.pos.latitude * (double)1e7); state.have_undulation = true; state.undulation = _buffer.pos.altitude_msl - _buffer.pos.altitude_ellipsoid; set_alt_amsl_cm(state, _buffer.pos.altitude_msl * 100); state.status = next_fix; _new_position = true; </pre>	Receiver backend writes GPS_State.location/status from the Emlid Reach ERB protocol. One of 14 receiver backends, each catalogued individually. [SHARED-STRUCT] target GPS_State.
9	libraries/AP_GPS/AP_GPS_MAV.cpp	67-75	AP_GPS_MAV::handle_message()	Source	<pre> state.status = (AP_GPS::GPS_Status)packet.fix_type; Location loc = {}; loc.lat = packet.lat; loc.lng = packet.lon; if (have_alt) { loc.alt = packet.alt * 100; // convert to centimeters } state.location = loc; </pre>	Receiver backend writes GPS_State.location/status from the MAVLink GPS_INPUT message. One of 14 receiver backends, each catalogued individually. [SHARED-STRUCT] target GPS_State.
10	libraries/AP_GPS/AP_GPS_MSP.cpp	44-51	AP_GPS_MSP::handle_msp()	Source	<pre> state.status = (AP_GPS::GPS_Status)pkt.fix_type; state.num_sats = pkt.satellites_in_view; state.location = { pkt.latitude, pkt.longitude, pkt.msl_altitude, Location::AltFrame::ABSOLUTE }; </pre>	Receiver backend writes GPS_State.location/status from the MSP protocol GPS data. One of 14 receiver backends, each catalogued individually. [SHARED-STRUCT] target GPS_State.
11	libraries/AP_GPS/AP_GPS_GSOF.cpp	231-238	AP_GPS_GSOF::pack_state_data()	Source	<pre> state.location.lat = (int32_t)(RAD_TO_DEG_DOUBLE * position.latitude_rad * (double)1e7); state.location.lng = (int32_t)(RAD_TO_DEG_DOUBLE * position.longitude_rad * (double)1e7); state.location.alt = (int32_t)(position.altitude * 100); state.last_gps_time_ms = AP_HAL::millis(); if ((vel.velocity_flags & 1) == 1) { state.ground_speed = vel.horizontal_velocity; state.ground_course = wrap_360(degrees(vel.heading)); } </pre>	ArduPilot_PNT_Reference_Audit.pdf Receiver backend writes GPS_State.location/status from the Trimble GSOF binary. One of 14 receiver backends, each catalogued individually. [SHARED-STRUCT] target GPS_State.

#	File Path	Line(s)	Function/Class	Role	Code Snippet (5-10 lines)	Notes
12	libraries/AP_GPS/AP_GPS_DroneCAN.cpp	388-396	AP_GPS_DroneCAN::handle_fix2_msg()	Source	<pre> if (msg.height_msl_mm != msg.height_ellipsoid_mm) { seen_valid_height_ellipsoid = true; } interim_state.have_undulation = seen_valid_height_ellipsoid; if (seen_valid_height_ellipsoid) { interim_state.undulation = (msg.height_msl_mm - msg.height_ellipsoid_mm) * 0.001; } interim_state.location = loc; set_alt_amsl_cm(interim_state, alt_amsl_cm); </pre>	Receiver backend writes GPS_State.location/status from the DroneCAN/UAVCAN Fix2 (external boundary modules/DroneCAN). One of 14 receiver backends, each catalogued individually. [SHARED-STRUCT] target GPS_State.
13	libraries/AP_GPS/AP_GPS_ExternalAHRS.cpp	49-56	AP_GPS_ExternalAHRS::handle_external()	Source	<pre> state.num_sats = pkt.satellites_in_view; const Location loc { pkt.latitude, pkt.longitude, pkt.msl_altitude, Location::AltFrame::ABSOLUTE }; </pre>	Receiver backend writes GPS_State.location/status from the External AHRS/INS GPS feed. One of 14 receiver backends, each catalogued individually. [SHARED-STRUCT] target GPS_State.
14	libraries/AP_GPS/AP_GPS_SITL.cpp	86-93	AP_GPS_SITL::read()	Source	<pre> state.status = AP_GPS::GPS_OK_FIX_3D; state.num_sats = 15; state.location = Location{ int32_t(latitude*1e7), int32_t(longitude*1e7), int32_t(altitude*100), Location::AltFrame::ABSOLUTE }; </pre>	Receiver backend writes GPS_State.location/status from the Software-in-the-loop simulated GPS. One of 14 receiver backends, each catalogued individually. [SHARED-STRUCT] target GPS_State.
15	libraries/AP_GPS/GPS_Backend.cpp	101-109	AP_GPS_Backend::fill_3d_velocity()	Transform	<pre> void AP_GPS_Backend::fill_3d_velocity(void) { float gps_heading = radians(state.ground_course); state.velocity.x = state.ground_speed * cosf(gps_heading); state.velocity.y = state.ground_speed * sinf(gps_heading); state.velocity.z = 0; state.have_vertical_velocity = false; } </pre>	Backend base class; derives state.velocity (NED) from ground_speed + ground_course for receivers that do not report 3D velocity. Shared by all backends.
16	libraries/AP_GPS/AP_GPS.h	324-329	AP_GPS::location()	Source	<pre> const Location &location(uint8_t instance) const { return state[instance].location; } const Location &location() const { return location(primary_instance); } </pre>	Frontend accessor exposing backend-measured GPS_State.location; dispatches to primary_instance. Reached vehicle-side via the AP::gps() singleton.
17	libraries/AP_GPS/AP_GPS.h	360-365	AP_GPS::velocity()	Source	<pre> const Vector3f &velocity(uint8_t instance) const { return state[instance].velocity; } const Vector3f &velocity() const { return velocity(primary_instance); } </pre>	3D NED velocity accessor of measured GPS_State.velocity. [MULTI-CATEGORY] velocity also drives Navigation fusion (Table 2).
18	libraries/AP_GPS/AP_GPS.h	192-201	struct AP_GPS::GPS_State	Source	<pre> struct GPS_State { uint8_t instance; // the instance number of this GPS // all the following fields must all be filled by the backend driver GPS_Status status; ///< driver fix status uint32_t time_week_ms; ///< GPS time (milliseconds from start of GPS week) uint16_t time_week; ///< GPS week number Location location; ///< last fix location float ground_speed; ///< ground speed in m/s float ground_course; ///< ground course in degrees, wrapped 0-360 </pre>	ArduPilot_PNT_Reference_Audit.pdf [SHARED-STRUCT][MULTI-CATEGORY] One struct mixes Positioning (location), Navigation (velocity, ground_speed, ground_course) and Timing (time_week_ms, time_week) — see Table 3 #8. Primary extraction-risk struct for a PNT split.
19	libraries/AP_GPS/AP_GPS_Blanded.cpp	17-25	AP_GPS_Blanded::calc_weights()	Transform	<pre> bool AP_GPS_Blanded::calc_weights(void) { // zero the blend weights memset(&blend_weights, 0, sizeof(blend_weights)); static_assert(GPS_MAX_RECEIVERS == 2, "GPS blending only currently works with 2 receivers"); // Note that the early quit below relies upon exactly 2 instances // The time delta calculations below also rely upon every instance being currently detected and being parsed </pre>	Multi-receiver blending of location/velocity (GPS_MAX_RECEIVERS==2). Reads gps.state[].status (Data dependency) to weight receivers.

#	File Path	Line(s)	Function/Class	Role	Code Snippet (5-10 lines)	Notes
20	libraries/AP_Common/Location.h	11-20	class Location	Source	<pre>uint8_t relative_alt : 1; // 1 if altitude is relative to home uint8_t loiter_ccw : 1; // 0 if clockwise, 1 if counter clockwise uint8_t terrain_alt : 1; // this altitude is above terrain uint8_t origin_alt : 1; // this altitude is above ekf origin uint8_t loiter_xtrack : 1; // 0 to crosstrack from center of waypoint, 1 to crosstrack from tangent exit location // note that mission storage only stores 24 bits of altitude (~ +/- 83km) int32_t alt; // in cm int32_t lat; // in 1E7 degrees int32_t lng; // in 1E7 degrees</pre>	[SHARED-STRUCT] Geodetic PNT fields lat/lng (1E7 deg) and alt (cm) packed with NON-PNT bitfields relative_alt, loiter_ccw, terrain_alt, origin_alt, loiter_xtrack. Ubiquitous value type; top extraction risk.
21	libraries/AP_Common/Location.cpp	259-268	Location::get_vector_from_origin_NEU_cm()	Transform	<pre>bool Location::get_vector_from_origin_NEU_cm(T &vec_neu) const { // convert altitude int32_t alt_above_origin_cm = 0; if (!get_alt_cm(AltFrame::ABOVE_ORIGIN, alt_above_origin_cm)) { return false; } // convert lat, lon if (!get_vector_xy_from_origin_NE_cm(vec_neu.xy())) {</pre>	Geodetic (lat/lng/alt) NEU centimetre vector relative to → the EKF origin (geodetic↔NED duality).
22	libraries/AP_InertialNav/AP_InertialNav.cpp	14-23	AP_InertialNav::update()	Transform	<pre>void AP_InertialNav::update(bool high_vibes) { // get the NE position relative to the local earth frame origin Vector2f posNE; if (_ahrs_ekf.get_relative_position_NE_origin_float(posNE)) { _relpos_cm.x = posNE.x * 100; // convert from m to cm _relpos_cm.y = posNE.y * 100; // convert from m to cm } // get the D position relative to the local earth frame origin</pre>	Pulls EKF origin-relative NE/D position into _relpos_cm (cm). get_position_neu_cm() exposes it to vehicle code.
23	libraries/AC_AttitudeControl/AC_PosControl.cpp	363-372	AC_PosControl::input_pos_NEU_cm()	Sink	<pre>void AC_PosControl::input_pos_NEU_cm(const Vector3p& pos_neu_cm, float pos_terrain_target_u_cm, float terrain_buffer_cm) { input_pos_NEU_m(pos_neu_cm * 0.01, pos_terrain_target_u_cm * 0.01, terrain_buffer_cm * 0.01); // Sets a new NEU position target in meters and computes a jerk-limited trajectory. // Updates internal acceleration commands using a smooth kinematic path constrained // by configured acceleration and jerk limits. Terrain margin is used to constrain // horizontal velocity to avoid vertical buffer violation. void AC_PosControl::input_pos_NEU_m(const Vector3p& pos_neu_m, float pos_terrain_target_u_m, float terrain_buffer_m)</pre>	ArduPilot_PNT_Reference_Audit.pdf 3D position/velocity controller consuming the position target. [ABSENT] there is no standalone AC_PosControl library — it physically resides under libraries/AC_AttitudeControl/ (snapshot-fidelity note).
24	libraries/APM_Control/AR_PosControl.cpp	114-123	AR_PosControl::update()	Sink	<pre>void AR_PosControl::update(float dt) { // exit immediately if no current location, destination or disarmed Vector3p curr_pos_NED_m; Vector3f curr_vel_NED; if (!hal.util->get_soft_armed() !AP::ahrs().get_relative_position_NED_origin(curr_pos_NED_m) !AP::ahrs().get_velocity_NED(curr_vel_NED)) { _desired_speed = _atc.get_desired_speed_accel_limited(0.0f, dt); _desired_lat_accel = 0.0f; _desired_turn_rate_rads = 0.0f; }</pre>	Rover position controller; reads AP::ahrs().get_relative_position_NED_origin() and get_velocity_NED() (Shared global/singleton). Separate controller from Copter's AC_PosControl.
25	libraries/AP_Beacon/AP_Beacon.cpp	175-184	AP_Beacon::get_vehicle_position_ned()	Source	<pre>bool AP_Beacon::get_vehicle_position_ned(Vector3f &position, float& accuracy_estimate) const { if (!device_ready()) { return false; } // check for timeout if (AP_HAL::millis() - veh_pos_update_ms > AP_BEACON_TIMEOUT_MS) { return false; } }</pre>	Non-GPS (GPS-denied) positioning source; returns vehicle NED position with a staleness timeout.

#	File Path	Line(s)	Function/Class	Role	Code Snippet (5-10 lines)	Notes
26	libraries/AP_VisualOdom/AP_VisualOdom.cpp	201-210	AP_VisualOdom::handle_pose_estimate()	Source	<pre>void AP_VisualOdom::handle_pose_estimate(uint64_t remote_time_us, uint32_t time_ms, float x, float y, float z, float roll, float pitch, float yaw, float posErr, float angErr, uint8_t reset_counter, int8_t quality) { // exit immediately if not enabled if (!enabled()) { return; } // call backend if (_driver != nullptr) { // convert attitude to quaternion and call backend } }</pre>	Non-GPS visual-odometry position/attitude source feeding EKF external-nav fusion.

Table 1a — Positioning dependencies (Layer 1 direct edges + Layer 2 transitive chains)**Layer 1 — Direct calls / called-by with typed dependency**

Ref #	Component/Function	Directly Calls	Directly Called By	Dependency Type	Code Snippet
1	AP_GPS_UBLOX backend	writes GPS_State.location/status; fill_3d_velocity()	AP_GPS::update() backend read() →	Inheritance / Interface	<pre>_last_pos_time = _buffer.posllh.itow; state.location.lng = _buffer.posllh.longitude; state.location.lat = _buffer.posllh.latitude; state.have_undulation = true; state.undulation = (_buffer.posllh.altitude_msl - _buffer.posllh.altitude_ellipsoid) * 0.001; set_alt_amsl_cm(state, _buffer.posllh.altitude_msl / 10); state.status = next_fix; _new_position = true;</pre>
2	AP_GPS_NMEA backend	writes GPS_State.location/status; fill_3d_velocity()	AP_GPS::update() backend read() →	Inheritance / Interface	<pre>state.location.lat = _new_latitude; state.location.lng = _new_longitude; } state.num_sats = _new_satellite_count; state.hdop = _new_hdop; switch(_new_quality_indicator) { case 0: // Fix not available or invalid state.status = AP_GPS:NO_FIX;</pre>
3	AP_GPS_SBF backend	writes GPS_State.location/status; fill_3d_velocity()	AP_GPS::update() backend read() →	Inheritance / Interface	<pre>state.location.lat = (int32_t)(temp.Latitude * RAD_TO_DEG_DOUBLE * (double)1e7); state.location.lng = (int32_t)(temp.Longitude * RAD_TO_DEG_DOUBLE * (double)1e7); state.have_undulation = !is_DNU(temp.Undulation); double height = temp.Height; // in metres if (state.have_undulation) { height -= temp.Undulation; state.undulation = -temp.Undulation; }</pre>
4	AP_GPS_NOVA backend	writes GPS_State.location/status; fill_3d_velocity()	AP_GPS::update() backend read() →	Inheritance / Interface	<pre>state.last_gps_time_ms = AP_HAL::millis(); state.location.lat = (int32_t) (bestposu.lat * (double)1e7); state.location.lng = (int32_t) (bestposu.lng * (double)1e7); state.have_undulation = true; state.undulation = -bestposu.undulation; set_alt_amsl_cm(state, bestposu.hgt * 100);</pre>
5	AP_GPS_SBP backend	writes GPS_State.location/status; fill_3d_velocity()	AP_GPS::update() backend read() →	Inheritance / Interface	<pre>state.location.lat = (int32_t) (pos_llh->lat * (double)1e7); state.location.lng = (int32_t) (pos_llh->lon * (double)1e7); state.location.alt = (int32_t) (pos_llh->height * 100); state.num_sats = pos_llh->n_sats; if (pos_llh->flags == 0) { state.status = AP_GPS::GPS_OK_FIX_3D;</pre>
6	AP_GPS_SBP2 backend	writes GPS_State.location/status; fill_3d_velocity()	AP_GPS::update() backend read() →	Inheritance / Interface	<pre>// state.location.lat = (int32_t) (last_pos_llh.lat * (double)1e7); state.location.lng = (int32_t) (last_pos_llh.lon * (double)1e7); state.location.alt = (int32_t) (last_pos_llh.height * 100); state.num_sats = last_pos_llh.n_sats; switch (last_pos_llh.flags.fix_mode) { case 1:</pre>

Ref #	Component/Function	Directly Calls	Directly Called By	Dependency Type	Code Snippet
7	AP_GPS_SIRF backend	writes GPS_State.location/status; fill_3d_velocity()	AP_GPS::update() backend read() →	Inheritance / Interface	<pre> state.status = AP_GPS::NO_FIX; }else if ((_buffer.nav.fix_type & FIX_MASK) == FIX_3D) { state.status = AP_GPS::GPS_OK_FIX_3D; }else{ state.status = AP_GPS::GPS_OK_FIX_2D; } state.location.lat = int32_t(be32toh(_buffer.nav.latitude)); state.location.lng = int32_t(be32toh(_buffer.nav.longitude)); </pre>
8	AP_GPS_ERB backend	writes GPS_State.location/status; fill_3d_velocity()	AP_GPS::update() backend read() →	Inheritance / Interface	<pre> _last_pos_time = _buffer.pos.time; state.location.lng = (int32_t)(_buffer.pos.longitude * (double)1e7); state.location.lat = (int32_t)(_buffer.pos.latitude * (double)1e7); state.have_undulation = true; state.undulation = _buffer.pos.altitude_msl - _buffer.pos.altitude_ellipsoid; set_alt_amsl_cm(state, _buffer.pos.altitude_msl * 100); state.status = next_fix; _new_position = true; </pre>
9	AP_GPS_MAV backend	writes GPS_State.location/status; fill_3d_velocity()	AP_GPS::update() backend read() →	Inheritance / Interface	<pre> state.status = (AP_GPS::GPS_Status)packet.fix_type; Location loc = {}; loc.lat = packet.lat; loc.lng = packet.lon; if (have_alt) { loc.alt = packet.alt * 100; // convert to centimeters } state.location = loc; </pre>
10	AP_GPS_MSP backend	writes GPS_State.location/status; fill_3d_velocity()	AP_GPS::update() backend read() →	Inheritance / Interface	<pre> state.status = (AP_GPS::GPS_Status)pkt.fix_type; state.num_sats = pkt.satellites_in_view; state.location = { pkt.latitude, pkt.longitude, pkt.msl_altitude, Location::AltFrame::ABSOLUTE }; </pre>
11	AP_GPS_GSOFF backend	writes GPS_State.location/status; fill_3d_velocity()	AP_GPS::update() backend read() →	Inheritance / Interface	<pre> state.location.lat = (int32_t)(RAD_TO_DEG_DOUBLE * position.latitude_rad * (double)1e7); state.location.lng = (int32_t)(RAD_TO_DEG_DOUBLE * position.longitude_rad * (double)1e7); state.location.alt = (int32_t)(position.altitude * 100); state.last_gps_time_ms = AP_HAL::millis(); if ((vel.velocity_flags & 1) == 1) { state.ground_speed = vel.horizontal_velocity; state.ground_course = wrap_360(degrees(vel.heading)); } </pre>
12	AP_GPS_DroneCAN backend	writes GPS_State.location/status; fill_3d_velocity()	AP_GPS::update() backend read() →	Inheritance / Interface	<pre> if (msg.height_msl_mm != msg.height_ellipsoid_mm) { seen_valid_height_ellipsoid = true; } interim_state.have_undulation = seen_valid_height_ellipsoid; if (seen_valid_height_ellipsoid) { interim_state.undulation = (msg.height_msl_mm - msg.height_ellipsoid_mm) * 0.001; } interim_state.location = loc; set_alt_amsl_cm(interim_state, alt_amsl_cm); </pre>
13	AP_GPS_ExternalAHRS backend	writes GPS_State.location/status; fill_3d_velocity()	AP_GPS::update() backend read() →	Inheritance / Interface	<pre> state.num_sats = pkt.satellites_in_view; const Location loc { pkt.latitude, pkt.longitude, pkt.msl_altitude, Location::AltFrame::ABSOLUTE }; </pre>

Ref #	Component/Function	Directly Calls	Directly Called By	Dependency Type	Code Snippet
14	AP_GPS_SITL backend	writes GPS_State.location/status; fill_3d_velocity()	AP_GPS::update() backend read() →	Inheritance / Interface	<pre> state.status = AP_GPS::GPS_OK_FIX_3D; state.num_sats = 15; state.location = Location{ int32_t(latitude*1e7), int32_t(longitude*1e7), int32_t(altitude*100), Location::AltFrame::ABSOLUTE </pre>
15	AP_GPS_Backend::fill_3d_velocity ()	cosf/sinf on state.ground_course; writes state.velocity	All receiver backends (UBLOX, NMEA, SIRF, ...)	Data dependency	<pre> void AP_GPS_Backend::fill_3d_velocity(void) { float gps_heading = radians(state.ground_course); state.velocity.x = state.ground_speed * cosf(gps_heading); state.velocity.y = state.ground_speed * sinf(gps_heading); state.velocity.z = 0; state.have_vertical_velocity = false; } </pre>
16	AP_GPS::location()	returns state[instance].location	NavEKF3_Measurements::readGpsData (); AP_AHRS; AC_WPNV	Function call	<pre> const Location &location(uint8_t instance) const { return state[instance].location; } const Location &location() const { return location(primary_instance); } </pre>
17	AP_GPS::velocity()	returns state[instance].velocity	NavEKF3_Measurements::readGpsData () (PosVel fusion)	Function call	<pre> const Vector3f &velocity(uint8_t instance) const { return state[instance].velocity; } const Vector3f &velocity() const { return velocity(primary_instance); } </pre>
18	AP_GPS::GPS_State (struct)	(data struct — no calls)	All GPS backends (write); frontend accessors (read)	Data dependency	<pre> struct GPS_State { uint8_t instance; // the instance number of this GPS // all the following fields must all be filled by the backend driver GPS_Status status; ///< driver fix status uint32_t time_week_ms; ///< GPS time (milliseconds from start of GPS week) uint16_t time_week; ///< GPS week number Location location; ///< last fix location float ground_speed; ///< ground speed in m/s float ground_course; ///< ground course in degrees, wrapped 0-360 </pre>
19	AP_GPS_Blended::_calc_weights()	gps.state[].status; time deltas	AP_GPS_Blended::_calc_weights()/update (blended primary)	Data dependency	<pre> bool AP_GPS_Blended::_calc_weights(void) { // zero the blend weights memset(&blend_weights, 0, sizeof(blend_weights)); static_assert(GPS_MAX_RECEIVERS == 2, "GPS blending only currently works with 2 receivers"); // Note that the early quit below relies upon exactly 2 instances // The time delta calculations below also rely upon every instance being currently detected and being parsed </pre>
20	class Location	get_alt_cm(); get_vector_xy_from_origin_NE_cm()	GPS_State; AP_AHRS; AC_WPNV; AP_Mission (ubiquitous)	Data dependency	<pre> uint8_t relative_alt : 1; // 1 if altitude is relative to home uint8_t loiter_ccw : 1; // 0 if clockwise, 1 if counter clockwise uint8_t terrain_alt : 1; // this altitude is above terrain uint8_t origin_alt : 1; // this altitude is above ekf origin uint8_t loiter_xtrack : 1; // 0 to crosstrack from center of waypoint, 1 to crosstrack from tangent exit location // note that mission storage only stores 24 bits of altitude (~ +/- 83km) int32_t alt; // in cm int32_t lat; // in 1E7 degrees int32_t lng; // in 1E7 degrees </pre>

Ref #	Component/Function	Directly Calls	Directly Called By	Dependency Type	Code Snippet
21	Location::get_vector_from_origin_NEU_cm()	get_alt_cm(); get_vector_xy_from_origin_NE_cm()	AC_WPNav::get_vector_NEU_cm()	Function call	<pre> bool Location::get_vector_from_origin_NEU_cm(T &vec_neu) const { // convert altitude int32_t alt_above_origin_cm = 0; if (!get_alt_cm(AltFrame::ABOVE_ORIGIN, alt_above_origin_cm)) { return false; } // convert lat, lon if (!get_vector_xy_from_origin_NE_cm(vec_neu.xy())) { </pre>
22	AP_InertialNav::update()	_ahrs_ekf.get_relative_position_NE_origin_float()	Copter::read_inertia()	Function call	<pre> void AP_InertialNav::update(bool high_vibes) { // get the NE position relative to the local earth frame origin Vector2f posNE; if (_ahrs_ekf.get_relative_position_NE_origin_float(posNE)) { _relpos_cm.x = posNE.x * 100; // convert from m to cm _relpos_cm.y = posNE.y * 100; // convert from m to cm } // get the D position relative to the local earth frame origin </pre>
23	AC_PosControl::input_pos_NEU_cm()	input_pos_NEU_m()	ArduCopter mode_auto.cpp / mode.cpp	Function call	<pre> void AC_PosControl::input_pos_NEU_cm(const Vector3p& pos_neu_cm, float pos_terrain_target_u_cm, float terrain_buffer_cm) { input_pos_NEU_m(pos_neu_cm * 0.01, pos_terrain_target_u_cm * 0.01, terrain_buffer_cm * 0.01); } // Sets a new NEU position target in meters and computes a jerk-limited trajectory. // Updates internal acceleration commands using a smooth kinematic path constrained // by configured acceleration and jerk limits. Terrain margin is used to constrain // horizontal velocity to avoid vertical buffer violation. void AC_PosControl::input_pos_NEU_m(const Vector3p& pos_neu_m, float pos_terrain_target_u_m, float terrain_buffer_m) </pre>
24	AR_PosControl::update()	AP::ahrs().get_relative_position_NED_origin(); get_velocity_NED()	Rover mode controllers	Shared global / singleton	<pre> void AR_PosControl::update(float dt) { // exit immediately if no current location, destination or disarmed Vector3p curr_pos_NED_m; Vector3f curr_vel_NED; if (!hal.util->get_soft_armed() !AP::ahrs().get_relative_position_NED_origin(curr_pos_NED_m) !AP::ahrs().get_velocity_NED(curr_vel_NED)) { _desired_speed = _atc.get_desired_speed_accel_limited(0.0f, dt); _desired_lat_accel = 0.0f; _desired_turn_rate_rads = 0.0f; } </pre>
25	AP_Beacon::get_vehicle_position_ned()	device_ready(); AP_HAL::millis()	NavEKF3 range-beacon fusion	Function call	<pre> bool AP_Beacon::get_vehicle_position_ned(Vector3f &position, float& accuracy_estimate) const { if (!device_ready()) { return false; } // check for timeout if (AP_HAL::millis() - veh_pos_update_ms > AP_BEACON_TIMEOUT_MS) { return false; } </pre>
26	AP_VisualOdom::handle_pose_estimate()	_driver backend handle_pose_estimate()	GCS / AP_ExternalAHRS pose handlers	Inheritance / Interface	<pre> void AP_VisualOdom::handle_pose_estimate(uint64_t remote_time_us, uint32_t time_ms, float x, float y, float z, float roll, float pitch, float yaw, float posErr, float angErr, uint8_t reset_counter, int8_t quality) { // exit immediately if not enabled if (!enabled()) { return; } // call backend if (_driver != nullptr) { // convert attitude to quaternion and call backend } </pre>

Layer 2 — Full transitive chain to final consumer (hop count)

Ref #	PNT Origin	Full Dependency Chain	Final Consumer	Chain Depth	Notes
1	AP_GPS_UBLOX (state.location)	AP_GPS_UBLOX AP_GPS::location()/velocity() NavEKF3_Measurements::readGpsData → → (gps.location/gps.velocity) NavEKF3::FuseVelPosNED AP_AHRS::get_location() → → → AC_WPNav AC_PosControl AC_AttitudeControl AP_Motors → → →	AP_Motors / SRV_Channel	8	Sensor source to actuator; GPS ingested in AP_NavEKF3_Measurements.cpp (readGpsData) before FuseVelPosNED.
2	AP_GPS_NMEA (state.location)	AP_GPS_NMEA AP_GPS::location()/velocity() NavEKF3_Measurements::readGpsData → → (gps.location/gps.velocity) NavEKF3::FuseVelPosNED AP_AHRS::get_location() → → → AC_WPNav AC_PosControl AC_AttitudeControl AP_Motors → → →	AP_Motors / SRV_Channel	8	Sensor source to actuator; GPS ingested in AP_NavEKF3_Measurements.cpp (readGpsData) before FuseVelPosNED.
3	AP_GPS_SBF (state.location)	AP_GPS_SBF AP_GPS::location()/velocity() NavEKF3_Measurements::readGpsData → → (gps.location/gps.velocity) NavEKF3::FuseVelPosNED AP_AHRS::get_location() → → → AC_WPNav AC_PosControl AC_AttitudeControl AP_Motors → → →	AP_Motors / SRV_Channel	8	Sensor source to actuator; GPS ingested in AP_NavEKF3_Measurements.cpp (readGpsData) before FuseVelPosNED.
4	AP_GPS_NOVA (state.location)	AP_GPS_NOVA AP_GPS::location()/velocity() NavEKF3_Measurements::readGpsData → → (gps.location/gps.velocity) NavEKF3::FuseVelPosNED AP_AHRS::get_location() → → → AC_WPNav AC_PosControl AC_AttitudeControl AP_Motors → → →	AP_Motors / SRV_Channel	8	Sensor source to actuator; GPS ingested in AP_NavEKF3_Measurements.cpp (readGpsData) before FuseVelPosNED.
5	AP_GPS_SBP (state.location)	AP_GPS_SBP AP_GPS::location()/velocity() NavEKF3_Measurements::readGpsData → → (gps.location/gps.velocity) NavEKF3::FuseVelPosNED AP_AHRS::get_location() → → → AC_WPNav AC_PosControl AC_AttitudeControl AP_Motors → → →	AP_Motors / SRV_Channel	8	Sensor source to actuator; GPS ingested in AP_NavEKF3_Measurements.cpp (readGpsData) before FuseVelPosNED.
6	AP_GPS_SBP2 (state.location)	AP_GPS_SBP2 AP_GPS::location()/velocity() NavEKF3_Measurements::readGpsData → → (gps.location/gps.velocity) NavEKF3::FuseVelPosNED AP_AHRS::get_location() → → → AC_WPNav AC_PosControl AC_AttitudeControl AP_Motors → → →	AP_Motors / SRV_Channel	8	Sensor source to actuator; GPS ingested in AP_NavEKF3_Measurements.cpp (readGpsData) before FuseVelPosNED.
7	AP_GPS_SIRF (state.location)	AP_GPS_SIRF AP_GPS::location()/velocity() NavEKF3_Measurements::readGpsData → → (gps.location/gps.velocity) NavEKF3::FuseVelPosNED AP_AHRS::get_location() → → → AC_WPNav AC_PosControl AC_AttitudeControl AP_Motors → → →	AP_Motors / SRV_Channel	8	Sensor source to actuator; GPS ingested in AP_NavEKF3_Measurements.cpp (readGpsData) before FuseVelPosNED.
8	AP_GPS_ERB (state.location)	AP_GPS_ERB AP_GPS::location()/velocity() NavEKF3_Measurements::readGpsData → → (gps.location/gps.velocity) NavEKF3::FuseVelPosNED AP_AHRS::get_location() → → → AC_WPNav AC_PosControl AC_AttitudeControl AP_Motors → → →	AP_Motors / SRV_Channel	8	Sensor source to actuator; GPS ingested in AP_NavEKF3_Measurements.cpp (readGpsData) before FuseVelPosNED.
9	AP_GPS_MAV (state.location)	AP_GPS_MAV AP_GPS::location()/velocity() NavEKF3_Measurements::readGpsData → → (gps.location/gps.velocity) NavEKF3::FuseVelPosNED AP_AHRS::get_location() → → → AC_WPNav AC_PosControl AC_AttitudeControl AP_Motors → → →	AP_Motors / SRV_Channel	8	Sensor source to actuator; GPS ingested in AP_NavEKF3_Measurements.cpp (readGpsData) before FuseVelPosNED.
10	AP_GPS_MSP (state.location)	AP_GPS_MSP AP_GPS::location()/velocity() NavEKF3_Measurements::readGpsData → → (gps.location/gps.velocity) NavEKF3::FuseVelPosNED AP_AHRS::get_location() → → → AC_WPNav AC_PosControl AC_AttitudeControl AP_Motors → → →	AP_Motors / SRV_Channel	8	Sensor source to actuator; GPS ingested in AP_NavEKF3_Measurements.cpp (readGpsData) before FuseVelPosNED.
11	AP_GPS_GSOF (state.location)	AP_GPS_GSOF AP_GPS::location()/velocity() NavEKF3_Measurements::readGpsData → → (gps.location/gps.velocity) NavEKF3::FuseVelPosNED AP_AHRS::get_location() → → → AC_WPNav AC_PosControl AC_AttitudeControl AP_Motors → → →	AP_Motors / SRV_Channel	8	Sensor source to actuator; GPS ingested in AP_NavEKF3_Measurements.cpp (readGpsData) before FuseVelPosNED.
12	AP_GPS_DroneCAN (state.location)	AP_GPS_DroneCAN AP_GPS::location()/velocity() NavEKF3_Measurements::readGpsData → → (gps.location/gps.velocity) NavEKF3::FuseVelPosNED AP_AHRS::get_location() → → → AC_WPNav AC_PosControl AC_AttitudeControl AP_Motors → → →	AP_Motors / SRV_Channel	8	Sensor source to actuator; GPS ingested in AP_NavEKF3_Measurements.cpp (readGpsData) before FuseVelPosNED.
13	AP_GPS_ExternalAHRS (state.location)	AP_GPS_ExternalAHRS AP_GPS::location()/velocity() → → NavEKF3_Measurements::readGpsData (gps.location/gps.velocity) NavEKF3::FuseVelPosNED → AP_AHRS::get_location() AC_WPNav AC_PosControl AC_AttitudeControl AP_Motors → → → →	AP_Motors / SRV_Channel	8	Sensor source to actuator; GPS ingested in AP_NavEKF3_Measurements.cpp (readGpsData) before FuseVelPosNED.
14	AP_GPS_SITL (state.location)	AP_GPS_SITL AP_GPS::location()/velocity() NavEKF3_Measurements::readGpsData → → (gps.location/gps.velocity) NavEKF3::FuseVelPosNED AP_AHRS::get_location() → → → AC_WPNav AC_PosControl AC_AttitudeControl AP_Motors → → →	AP_Motors / SRV_Channel	8	Sensor source to actuator; GPS ingested in AP_NavEKF3_Measurements.cpp (readGpsData) before FuseVelPosNED.
15	GPS velocity (fill_3d_velocity)	GPS_State.velocity AP_GPS::velocity() NavEKF3_Measurements::readGpsData → → → NavEKF3::FuseVelPosNED AP_AHRS::get_velocity_NED() AC_PosControl AP_Motors → → →	AP_Motors	6	[MULTI-CATEGORY] also Navigation.
16	AP_GPS::location()	AP_GPS::location() NavEKF3_Measurements::readGpsData (gps.location) → → NavEKF3::FuseVelPosNED AP_AHRS::get_location() AC_WPNav::set_wp_destination_loc → → → AC_PosControl AP_Motors →	AP_Motors	6	ArduPilot_PNT_Reference_Audit.pdf Geodetic position path (ingestion hop explicit).
17	AP_GPS::velocity()	AP_GPS::velocity() NavEKF3_Measurements::readGpsData (gps.velocity) → → NavEKF3::FuseVelPosNED AP_AHRS::get_velocity_NED() AC_PosControl AP_Motors → → →	AP_Motors	5	Velocity path (ingestion hop explicit).
18	GPS_State (struct)	GPS_State AP_GPS accessors {location velocity time_week} NavEKF3_Measurements → → → EKF/AHRS + AP_RTC controllers + clock →	AP_Motors and AP_RTC	4	[SHARED-STRUCT][MULTI-CATEGORY] branches into Table 3.
19	AP_GPS_Blended	Blended AP_GPS primary (blended) location/velocity NavEKF3_Measurements → → → NavEKF3::FuseVelPosNED AP_AHRS controllers AP_Motors → → →	AP_Motors	6	Transform upstream of EKF ingestion.

Ref #	PNT Origin	Full Dependency Chain	Final Consumer	Chain Depth	Notes
20	class Location (geodetic value type)	Location AC_WPNav::set_wp_destination_loc get_vector_NEU_cm AC_PosControl → → → → AC_AttitudeControl AP_Motors →	AP_Motors	5	[SHARED-STRUCT] Pervasive value type; depth measured along its dominant waypoint-control consumer branch (the AC_WPNav→PosControl path).
21	Location→NEU conversion	get_vector_from_origin_NEU_cm() AC_WPNav::get_vector_NEU_cm → → set_wp_destination_NEU_cm AC_PosControl →	AC_PosControl	3	Geodetic↔NED transform edge.
22	AP_InertialNav	EKF origin-relative pos AP_InertialNav::update_relpos_cm Copter logging/reporting → → →	GCS / logger	3	Reporting branch.
23	AC_PosControl (Sink)	AP_AHRS pos AC_WPNav AC_PosControl AC_AttitudeControl AP_Motors → → → →	AP_Motors	4	Inbound chain to a Sink endpoint.
24	AR_PosControl (Sink, Rover)	AP_AHRS pos/vel AR_PosControl AR_AttitudeControl SRV_Channels → → →	SRV_Channels	3	Rover actuation endpoint.
25	AP_Beacon (Source)	AP_Beacon get_vehicle_position_ned NavEKF3 RngBcn fusion AP_AHRS AP_Motors → → → →	AP_Motors	4	GPS-denied source.
26	AP_VisualOdom (Source)	AP_VisualOdom NavEKF3 external-nav fusion AP_AHRS AP_Motors → → →	AP_Motors	3	ArduPilot_PNT_Reference_Audit.pdf GPS-denied source.

Table 2 — Core Navigation (AHRS hub, EKF fusion, waypoint/mission nav, fixed-wing guidance & energy, attitude sink)

#	File Path	Line(s)	Function/Class	Role	Code Snippet (5-10 lines)	Notes	New Service Location
1	libraries/AP_AHRS/AP_AHRS.cpp	3600-3604	AP_AHRS::get_location()	Transform	<pre>bool AP_AHRS::get_location(Location &loc) const { loc = state.location; return state.location_ok; }</pre>	Central navigation hub returns the fused Location from the active EKF. Reached via AP::ahrs(). [CIRCULAR] AHRS owns NavEKF2/NavEKF3 while the EKF core reads AHRS/sensor state back through AP_DAL — AHRS↔EKF mutual reference.	—
2	libraries/AP_AHRS/AP_AHRS.h	238-246	AP_AHRS::groundspeed()	Sink	<pre>// EKF has a better ground speed vector estimate const Vector2f &groundspeed_vector() const { return state.ground_speed_vec; } // return ground speed estimate in meters/second. Used by ground vehicles. float groundspeed(void) const { return state.ground_speed; } const Vector3f &get_accel_ef() const { return state.accel_ef; }</pre>	Read accessor returning the already-fused state.ground_speed (m/s). Performs no fusion and produces nothing from a measurement — classified Sink (read accessor), NOT Source.	—
3	libraries/AP_AHRS/AP_AHRS.h	269-277	AP_AHRS::get_origin() / have_inertial_nav()	Sink	<pre>// returns the inertial navigation origin in lat/lon/alt bool get_origin(Location &ret) const WARN_IF_UNUSED; bool have_inertial_nav() const; // return a ground velocity in meters/second, North/East/Down // order. Must only be called if have_inertial_nav() is true bool get_velocity_NED(Vector3f &vec) const WARN_IF_UNUSED;</pre>	Availability/origin read accessors (have_inertial_nav()) is a user-cited symbol, L273). They report EKF availability/origin — read accessors, NOT measurement Sources; classified Sink. EKF origin is WRITTEN by the EKF (see #15 calcGpsGoodToAlign / setOrigin).	—
4	libraries/AP_AHRS/AP_AHRS.cpp	1739-1748	AP_AHRS::get_relative_position_NED_origin()	Transform	<pre>bool AP_AHRS::get_relative_position_NED_origin(Vector3p &vec) const { switch (active_EKF_type()) { #if AP_AHRS_DCM_ENABLED case EKFType::DCM: return dcm.get_relative_position_NED_origin(vec); #endif #if HAL_NAVEKF2_AVAILABLE case EKFType::TWO: { Vector2p posNE;</pre>	Geodetic↔NED: origin-relative NED position via active-EKF-type dispatch (DCM/EKF2/EKF3) — performs conversion on read.	—
5	libraries/AP_AHRS/AP_AHRS.cpp	3662-3666	AP_AHRS::get_velocity_NED()	Transform	<pre>bool AP_AHRS::get_velocity_NED(Vector3f &vec) const { vec = state.velocity_NED; return state.velocity_NED_ok; }</pre>	Exposes the fused NED velocity (nav-hub output of EKF fusion); consumed by crash detection and Rover position control.	—

#	File Path	Line(s)	Function/Class	Role	Code Snippet (5-10 lines)	Notes	New Service Location
6	libraries/AP_NavEKF3/AP_NavEKF3_core.cpp	627-636	NavEKF3_core::UpdateFilter()	Transform	<pre>void NavEKF3_core::UpdateFilter(bool predict) { // don't run filter updates if states have not been initialised if (!statesInitialised) { return; } fill_scratch_variables(); // update sensor selection (for affinity)</pre>	ArduPilot_PNT_Reference_Audit.pdf Extended Kalman Filter predict/fuse step (the core PNT transform). [CIRCULAR] AHRS owns NavEKF2/NavEKF3 while the EKF core reads AHRS/sensor state back through AP_DAL — AHRS↔EKF mutual reference. One of 13 AP_NavEKF3 fusion modules, all catalogued.	—
7	libraries/AP_NavEKF3/AP_NavEKF3_PosVelFusion.cpp	502-511	NavEKF3_core::SelectVelPosFusion()	Transform	<pre>void NavEKF3_core::SelectVelPosFusion() { // Check if the magnetometer has been fused on that time step and the filter is running at faster than 200 Hz // If so, don't fuse measurements on this time step to reduce frame over-runs // Only allow one time slip to prevent high rate magnetometer data preventing fusion of other measurements if (magFusePerformed && dtIMUavg < 0.005f && !posVelFusionDelayed) { posVelFusionDelayed = true; return; } else { posVelFusionDelayed = false;</pre>	GPS position/velocity fusion scheduling; the dtIMUavg<0.005 gate couples EKF fusion cadence to the scheduler loop rate (timing coupling). [CIRCULAR] AHRS owns NavEKF2/NavEKF3 while the EKF core reads AHRS/sensor state back through AP_DAL — AHRS↔EKF mutual reference. One of 13 AP_NavEKF3 fusion modules, all catalogued.	—
8	libraries/AP_NavEKF3/AP_NavEKF3_PosVelFusion.cpp	694-702	NavEKF3_core::FuseVelPosNED()	Transform	<pre>void NavEKF3_core::FuseVelPosNED() { // declare variables used to control access to arrays bool fuseData[6] {}; uint8_t stateIndex; uint8_t obsIndex; // declare variables used by state and covariance update calculations Vector6 R_OBS; // Measurement variances used for fusion</pre>	Core position/velocity measurement-update (Kalman gain applied to GPS/baro NED innovations). One of 13 AP_NavEKF3 fusion modules, all catalogued.	—
9	libraries/AP_NavEKF3/AP_NavEKF3_Measurements.cpp	559-567	NavEKF3_core::readGpsData()	Transform	<pre>void NavEKF3_core::readGpsData() { // check for new GPS data const auto &gps = dal.gps(); // limit update rate to avoid overflowing the FIFO buffer if (gps.last_message_time_ms(selected_gps) - lastTimeGpsReceived_ms <= frontend->sensorIntervalMin_ms) { return; }</pre>	GPS ingestion into the EKF delayed-time buffer (reads gps.location()/velocity()/status via AP_DAL). This is the GPS→EKF hop omitted earlier (cf. Table 1a chains). [CIRCULAR] AHRS owns NavEKF2/NavEKF3 while the EKF core reads AHRS/sensor state back through AP_DAL — AHRS↔EKF mutual reference. One of 13 AP_NavEKF3 fusion modules, all catalogued.	—
10	libraries/AP_NavEKF3/AP_NavEKF3_Control.cpp	232-240	NavEKF3_core::setAidingMode()	Transform	<pre>void NavEKF3_core::setAidingMode() { resetDataSource posResetSource = resetDataSource::DEFAULT; resetDataSource velResetSource = resetDataSource::DEFAULT; // Save the previous status so we can detect when it has changed PV_AidingModePrev = PV_AidingMode; setYawSource();</pre>	Navigation aiding-mode state machine (none/relative/absolute) driving which PNT measurements are fused. One of 13 AP_NavEKF3 fusion modules, all catalogued.	—
11	libraries/AP_NavEKF3/AP_NavEKF3_MagFusion.cpp	227-235	NavEKF3_core::SelectMagFusion()	Transform	<pre>void NavEKF3_core::SelectMagFusion() { // clear the flag that lets other processes know that the expensive magnetometer fusion operation has been performed on that time step // used for load levelling magFusePerformed = false; // Store yaw angle when moving for use as a static reference when not moving if (!onGroundNotMoving) { if (fabsF(prevTnb[0][2]) < fabsF(prevTnb[1][2])) {</pre>	ArduPilot_PNT_Reference_Audit.pdf Magnetometer/yaw fusion scheduling feeding the heading component of the navigation solution. One of 13 AP_NavEKF3 fusion modules, all catalogued.	—

#	File Path	Line(s)	Function/Class	Role	Code Snippet (5-10 lines)	Notes	New Service Location
12	libraries/AP_NavEKF3/AP_NavEKF3_AirDataFusion.cpp	193-201	NavEKF3_core::SelectTasFusion()	Transform	<pre>void NavEKF3_core::SelectTasFusion() { // Check if the magnetometer has been fused on that time // step and the filter is running at faster than 200 Hz // If so, don't fuse measurements on this time step to // reduce frame over-runs // Only allow one time slip to prevent high rate // magnetometer data locking out fusion of other measurements if (magFusePerformed && dtIMUavg < 0.005f && !airSpdFusionDelayed) { airSpdFusionDelayed = true; return; } else {</pre>	True-airspeed / synthetic-sideslip fusion (wind/velocity estimation for fixed-wing nav). One of 13 AP_NavEKF3 fusion modules, all catalogued.	—
13	libraries/AP_NavEKF3/AP_NavEKF3_OptFlowFusion.cpp	21-29	NavEKF3_core::SelectFlowFusion()	Transform	<pre>void NavEKF3_core::SelectFlowFusion() { // Check if the magnetometer has been fused on that time // step and the filter is running at faster than 200 Hz // If so, don't fuse measurements on this time step to // reduce frame over-runs // Only allow one time slip to prevent high rate // magnetometer data preventing fusion of other measurements if (magFusePerformed && dtIMUavg < 0.005f && !optFlowFusionDelayed) { optFlowFusionDelayed = true; return; } else {</pre>	Optical-flow fusion scheduling for GPS-denied velocity/position aiding. One of 13 AP_NavEKF3 fusion modules, all catalogued.	—
14	libraries/AP_NavEKF3/AP_NavEKF3_Outputs.cpp	326-334	NavEKF3_core::getLLH()	Transform	<pre>bool NavEKF3_core::getLLH(Location &loc) const { Location origin; if (getOriginLLH(origin)) { postype_t posD; if (getPosD_local(posD) && PV_AidingMode != AID_NONE) { // Altitude returned is an absolute altitude // relative to the WGS-84 spheroid loc.set_alt_cm(origin.alt - posD*100.0, Location::AltFrame::ABSOLUTE); if (filterStatus.flags.horiz_pos_abs filterStatus.flags.horiz_pos_rel) {</pre>	EKF position output: converts the filter state to a geodetic Location consumed by AP_AHRS. One of 13 AP_NavEKF3 fusion modules, all catalogued.	—
15	libraries/AP_NavEKF3/AP_NavEKF3_VehicleStatus.cpp	14-22	NavEKF3_core::calcGpsGoodToAlign()	Transform	<pre>void NavEKF3_core::calcGpsGoodToAlign(void) { if (inFlight && !finalInflightYawInit && assume_zero_sideslip() && !use_compass()) { // this is a special case where a plane has launched // without magnetometer // is now in the air and needs to align yaw to the GPS // and start navigating as soon as possible gpsGoodToAlign = true; return; }</pre>	GPS-quality gate deciding whether the GPS is good enough to align/aid the navigation solution. One of 13 AP_NavEKF3 fusion modules, all catalogued.	—
16	libraries/AP_NavEKF3/AP_NavEKF3_GyroBias.cpp	6-13	NavEKF3_core::resetGyroBias()	Transform	<pre>void NavEKF3_core::resetGyroBias(void) { stateStruct.gyro_bias.zero(); zeroRows(P,10,12); zeroCols(P,10,12); P[10][10] = sq(radians(0.5f * dtIMUavg)); P[11][11] = P[10][10];</pre>	Gyro-bias state reset within the inertial-navigation state vector. One of 13 AP_NavEKF3 fusion modules, all catalogued.	—
17	libraries/AP_NavEKF3/AP_NavEKF3_RngBcnFusion.cpp	95-103	NavEKF3_core::FuseRngBcn()	Transform	<pre>void NavEKF3_core::FuseRngBcn() { // declarations ftype pn; ftype pe; ftype pd; ftype bcn_pn; ftype bcn_pe; ftype bcn_pd;</pre>	ArduPilot_PNT_Reference_Audit.pdf Range-beacon (non-GPS) position fusion — consumes AP_Beacon ranges (Table 1 #25). One of 13 AP_NavEKF3 fusion modules, all catalogued.	—

#	File Path	Line(s)	Function/Class	Role	Code Snippet (5-10 lines)	Notes	New Service Location
18	libraries/AP_NavEKF3/AP_NavEKF3_Logging.cpp	15-23	NavEKF3_core::Log_Write_XKF1()	Sink	<pre>void NavEKF3_core::Log_Write_XKF1(uint64_t time_us) const { // Write first EKF packet Vector3f euler; Vector2p posNE; postype_t posD; Vector3f velNED; Vector3f gyroBias; float posDownDeriv;</pre>	Logs the fused navigation state (pos/vel/attitude) to the dataflash; reads PNT state, drives no control — a logging Sink. One of 13 AP_NavEKF3 fusion modules, all catalogued.	—
19	libraries/AP_NavEKF3/AP_NavEKF3.cpp	1846-1854	NavEKF3::getFilterStatus()	Transform	<pre>void NavEKF3::getFilterStatus(nav_filter_status &status) const { if (core) { core[primary].getFilterStatus(status); } else { memset(&status, 0, sizeof(status)); } }</pre>	EKF3 frontend: exposes the per-core nav_filter_status (position/velocity validity bits) to vehicle health logic (Group 2).	—
20	libraries/AP_NavEKF2/AP_NavEKF2_core.cpp	528-536	NavEKF2_core::UpdateFilter()	Transform	<pre>void NavEKF2_core::UpdateFilter(bool predict) { // Set the flag to indicate to the filter that the // front-end has given permission for a new state prediction // cycle to be started startPredictEnabled = predict; // don't run filter updates if states have not been // initialised if (!statesInitialised) { return; } }</pre>	Second-generation EKF predict/fuse step; selectable alternative to EKF3. [CIRCULAR] AHRS owns NavEKF2/NavEKF3 while the EKF core reads AHRS/sensor state back through AP_DAL — AHRS↔EKF mutual reference. [DUPLICATED] parallel EKF implementation to AP_NavEKF3 (extraction note).	—
21	libraries/AP_NavEKF2/AP_NavEKF2_PosVelFusion.cpp	560-568	NavEKF2_core::FuseVelPosNED()	Transform	<pre>void NavEKF2_core::FuseVelPosNED() { // health is set bad until test passed bool velHealth = false; // boolean true if velocity // measurements have passed innovation consistency check bool posHealth = false; // boolean true if position // measurements have passed innovation consistency check bool hgtHealth = false; // boolean true if height // measurements have passed innovation consistency check // declare variables used to check measurement errors Vector3F velInnov;</pre>	EKF2 GPS position/velocity measurement-update; structurally mirrors EKF3 FuseVelPosNED (#8). [DUPLICATED].	—
22	libraries/AP_NavEKF/EKFGSF_yaw.cpp	30-38	EKFGSF_yaw::update()	Transform	<pre>void EKFGSF_yaw::update(const Vector3F &delAng, const Vector3F &delVel, const ftype delAngDT, const ftype delVelDT, bool runEKF, ftype TAS) { // copy to class variables</pre>	Gaussian-Sum-Filter emergency yaw estimator (shared by EKF2/EKF3) providing a GPS-velocity-derived yaw used to recover the navigation solution.	—
23	libraries/AC_WPNav/AC_WPNav.cpp	245-254	AC_WPNav::set_wp_destination_loc()	Transform	<pre>bool AC_WPNav::set_wp_destination_loc(const Location& destination) { bool is_terrain_alt; Vector3f dest_neu_cm; // convert destination location to vector if (!get_vector_NEU_cm(destination, dest_neu_cm, is_terrain_alt)) { return false; } }</pre>	ArduPilot_PNT_Reference_Audit.pdf Waypoint navigation: converts a Location waypoint to a NEU target (get_vector_NEU_cm) relative to EKF origin.	—

#	File Path	Line(s)	Function/Class	Role	Code Snippet (5-10 lines)	Notes	New Service Location
24	libraries/AP_Mission/AP_Mission.cpp	552-561	AP_Mission::get_next_nav_cmd()	Transform	<pre> bool AP_Mission::get_next_nav_cmd(uint16_t start_index, Mission_Command& cmd) { // search until the end of the mission command list for (uint16_t cmd_index = start_index; cmd_index < (unsigned)_cmd_total; cmd_index++) { // get next command if (!get_next_cmd(cmd_index, cmd, false)) { // no more commands so return failure return false; } // if found a "navigation" command then return it </pre>	Mission-command storage reader + navigation-command sequencing: scans stored commands and returns the next NAV waypoint. NOT a PNT measurement Source — it sequences stored navigation targets (Transform).	—
25	libraries/AP_L1_Control/AP_L1_Control.cpp	226-234	AP_L1_Control::update_waypoint()	Transform	<pre> // Calculate L1 gain required for specified damping float K_L1 = 4.0f * _L1_damping * _L1_damping; // Get current position and velocity if (_ahrs.get_location(_current_loc) == false) { // if no GPS loc available, maintain last nav/target_bearing _data_is_stale = true; return; } </pre>	Fixed-wing lateral (L1) guidance consuming the AHRS navigation position via _ahrs.get_location() (returns early if unavailable). The same method also reads AP_HAL::micros() at L214 for the dt timing coupling.	AHRS shim get_location(), fed by set_state_ne(n, e, ...)
26	libraries/AP_L1_Control/AP_L1_Control.cpp	79-88	AP_L1_Control::nav_roll_cd()	Transform	<pre> int32_t AP_L1_Control::nav_roll_cd(void) const { float ret; /* formula can be obtained through equations of balanced spiral: liftForce * cos(roll) = gravityForce * cos(pitch); liftForce * sin(roll) = gravityForce * lateralAcceleration / gravityAcceleration; // as mass = gravityForce/gravityAcceleration see issue 24319 [https://github.com/ArduPilot/ardupilot/issues/24319] Multiplier 100.0f is for converting degrees to centidegrees Made changes to avoid zero division as proposed by Andrew Tridgell: https://github.com/ArduPilot/ardupilot/pull/24331#d iscussion_r1267798397 </pre>	Derives demanded roll (centi-deg) from the lateral-acceleration demand and AHRS pitch.	get_roll_deg() = nav_roll_cd()/100 → L1_GetRollDeg
27	libraries/AP_TECS/AP_TEC S.cpp	1259-1268	AP_TECS::update_pitch_throttle()	Transform	<pre> void AP_TECS::update_pitch_throttle(int32_t hgt_dem_cm, int32_t EAS_dem_cm, enum AP_FixedWing::FlightStage flight_stage, float distance_beyond_land_wp, int32_t ptchMinCO_cd, int16_t throttle_nudge, float hgt_afe, float load_factor, float pitch_trim_deg) { </pre>	Fixed-wing Total Energy Control (speed/altitude) producing pitch & throttle demands.	—
28	libraries/AC_AttitudeControl/AC_AttitudeControl.cpp	411-419	AC_AttitudeControl::input_euler_angle_roll_pitch_yaw_cd()	Sink	<pre> void AC_AttitudeControl::input_euler_angle_roll_pitch_yaw_cd(float euler_roll_angle_cd, float euler_pitch_angle_cd, float euler_yaw_angle_cd, bool slew_yaw) { // Convert from centidegrees on public interface to radians const float euler_roll_angle_rad = cd_to_rad(euler_roll_angle_cd); const float euler_pitch_angle_rad = cd_to_rad(euler_pitch_angle_cd); const float euler_yaw_angle_rad = cd_to_rad(euler_yaw_angle_cd); input_euler_angle_roll_pitch_yaw_rad(euler_roll_angle_rad , euler_pitch_angle_rad, euler_yaw_angle_rad, slew_yaw); } </pre>	ArduPilot_PNT_Reference_Audit.pdf Attitude/rate controller consuming the navigation demand; downstream of all navigation.	—

Table 2a — Navigation dependencies (Layer 1 direct edges + Layer 2 transitive chains)**Layer 1 — Direct calls / called-by with typed dependency**

Ref #	Component/Function	Directly Calls	Directly Called By	Dependency Type	Code Snippet
1	AP_AHRS::get_location()	returns state.location (fused by active EKF)	ArduCopter inertia.cpp; commands.cpp; GCS_MAVLink_Copter.cpp	Function call	<pre>bool AP_AHRS::get_location(Location &loc) const { loc = state.location; return state.location_ok; }</pre>
2	AP_AHRS::groundspeed()	returns state.ground_speed	Rover speed control; GCS reporting	Data dependency	<pre>// EKF has a better ground speed vector estimate const Vector2f &groundspeed_vector() const { return state.ground_speed_vec; } // return ground speed estimate in meters/second. Used by ground vehicles. float groundspeed(void) const { return state.ground_speed; } const Vector3f &get_accel_ef() const { return state.accel_ef; }</pre>
3	AP_AHRS::have_inertial_nav()/get_origin()	active_EKF_type(); EKF origin	Copter::ekf_has_absolute_position (system.cpp); ekf_check	Function call	<pre>// returns the inertial navigation origin in lat/lon/alt bool get_origin(Location &ret) const WARN_IF_UNUSED; bool have_inertial_nav() const; // return a ground velocity in meters/second, North/East/Down // order. Must only be called if have_inertial_nav() is true bool get_velocity_NED(Vector3f &vec) const WARN_IF_UNUSED;</pre>
4	AP_AHRS::get_relative_position_NED_origin()	dcm/EKF2/EKF3 get_relative_position_NED_origin()	AR_PosControl::update; AP_InertialNav::update	Inheritance / Interface	<pre>bool AP_AHRS::get_relative_position_NED_origin(Vector3p &vec) const { switch (active_EKF_type()) { #if AP_AHRS_DCM_ENABLED case EKFType::DCM: return dcm.get_relative_position_NED_origin(vec); #endif #if HAL_NAVEKF2_AVAILABLE case EKFType::TWO: { Vector2p posNE;</pre>
5	AP_AHRS::get_velocity_NED()	returns state.velocity_NED	Copter::crash_check; AR_PosControl	Data dependency	<pre>bool AP_AHRS::get_velocity_NED(Vector3f &vec) const { vec = state.velocity_NED; return state.velocity_NED_ok; }</pre>
6	NavEKF3_core::UpdateFilter()	fill_scratch_variables(); Select*Fusion(); CovariancePrediction()	NavEKF3::UpdateFilter() (frontend, per core)	Function call	<pre>void NavEKF3_core::UpdateFilter(bool predict) { // don't run filter updates if states have not been initialised if (!statesInitialised) { return; } fill_scratch_variables(); // update sensor selection (for affinity)</pre>
7	NavEKF3_core::SelectVelPosFusion()	readGpsData(); FuseVelPosNED()	NavEKF3_core::UpdateFilter()	Function call	<pre>void NavEKF3_core::SelectVelPosFusion() { // Check if the magnetometer has been fused on that time step and the filter is running at faster than 200 Hz // If so, don't fuse measurements on this time step to reduce frame over-runs // Only allow one time slip to prevent high rate magnetometer data preventing fusion of other measurements if (magFusePerformed && dtIMUavg < 0.005f && !posVelFusionDelayed) { posVelFusionDelayed = true; return; } else { posVelFusionDelayed = false;</pre>

Ref #	Component/Function	Directly Calls	Directly Called By	Dependency Type	Code Snippet
8	NavEKF3_core::FuseVelPosNED()	Kalman gain on NED pos/vel innovations; correct stateStruct	NavEKF3_core::SelectVelPosFusion()	Function call	<pre> void NavEKF3_core::FuseVelPosNED() { // declare variables used to control access to arrays bool fuseData[6] {}; uint8_t stateIndex; uint8_t obsIndex; // declare variables used by state and covariance update calculations Vector6 R_OBS; // Measurement variances used for fusion </pre>
9	NavEKF3_core::readGpsData()	gps.location(); gps.velocity(); gps.status() (via AP_DAL); storedGPS.push()	NavEKF3_core::SelectVelPosFusion()/UpdateFilter()	Function call	<pre> void NavEKF3_core::readGpsData() { // check for new GPS data const auto &gps = dal.gps(); // limit update rate to avoid overflowing the FIFO buffer if (gps.last_message_time_ms(selected_gps) - lastTimeGpsReceived_ms <= frontend->sensorIntervalMin_ms) { return; } </pre>
10	NavEKF3_core::setAidingMode()	checks gpsGoodToAlign, posTimeout; sets PV_AidingMode	NavEKF3_core::controlFilterModes()	Function call	<pre> void NavEKF3_core::setAidingMode() { resetDataSource posResetSource = resetDataSource::DEFAULT; resetDataSource velResetSource = resetDataSource::DEFAULT; // Save the previous status so we can detect when it has changed PV_AidingModePrev = PV_AidingMode; setYawSource(); </pre>
11	NavEKF3_core::SelectMagFusion()	FuseMagnetometer(); FuseEulerYaw()	NavEKF3_core::UpdateFilter()	Function call	<pre> void NavEKF3_core::SelectMagFusion() { // clear the flag that lets other processes know that the expensive magnetometer fusion operation has been performed on that time step // used for load levelling magFusePerformed = false; // Store yaw angle when moving for use as a static reference when not moving if (!onGroundNotMoving) { if (fabsF(prevTnb[0][2]) < fabsF(prevTnb[1][2])) { </pre>
12	NavEKF3_core::SelectTasFusion()	FuseAirspeed()	NavEKF3_core::UpdateFilter()	Function call	<pre> void NavEKF3_core::SelectTasFusion() { // Check if the magnetometer has been fused on that time step and the filter is running at faster than 200 Hz // If so, don't fuse measurements on this time step to reduce frame over-runs // Only allow one time slip to prevent high rate magnetometer data locking out fusion of other measurements if (magFusePerformed && dtIMUavg < 0.005f && !airSpdFusionDelayed) { return; } else { </pre>
13	NavEKF3_core::SelectFlowFusion()	EstimateTerrainOffset(); FuseOptFlow()	NavEKF3_core::UpdateFilter()	Function call	<pre> void NavEKF3_core::SelectFlowFusion() { // Check if the magnetometer has been fused on that time step and the filter is running at faster than 200 Hz // If so, don't fuse measurements on this time step to reduce frame over-runs // Only allow one time slip to prevent high rate magnetometer data preventing fusion of other measurements if (magFusePerformed && dtIMUavg < 0.005f && !optFlowFusionDelayed) { return; } else { </pre>

Ref #	Component/Function	Directly Calls	Directly Called By	Dependency Type	Code Snippet
14	NavEKF3_core::getLLH()	outputDataNew / EKF_origin; builds Location	NavEKF3::getLLH() (frontend) -> AP_AHRS	Function call	<pre> bool NavEKF3_core::getLLH(Location &loc) const { Location origin; if (getOriginLLH(origin)) { postype_t posD; if (getPosD_local(posD) && PV_AidingMode != AID_NONE) { // Altitude returned is an absolute altitude relative to the WGS-84 spheroid loc.set_alt_cm(origin.alt - posD*100.0, Location::AltFrame::ABSOLUTE); if (filterStatus.flags.horiz_pos_abs filterStatus.flags.horiz_pos_rel) { </pre>
15	NavEKF3_core::calcGpsGoodToAlign()	reads gps metrics; sets gpsGoodToAlign	NavEKF3_core::readGpsData()/setAidingMode()	Function call	<pre> void NavEKF3_core::calcGpsGoodToAlign(void) { if (inFlight && !finalInflightYawInit && assume_zero_sideslip() && !use_compass()) { // this is a special case where a plane has launched without magnetometer // is now in the air and needs to align yaw to the GPS and start navigating as soon as possible gpsGoodToAlign = true; return; } </pre>
16	NavEKF3_core::resetGyroBias()	zeroes stateStruct.gyro_bias; resets P	NavEKF3_core::InitialiseFilterBootstrap()	Function call	<pre> void NavEKF3_core::resetGyroBias(void) { stateStruct.gyro_bias.zero(); zeroRows(P,10,12); zeroCols(P,10,12); P[10][10] = sq(radians(0.5f * dtIMUavg)); P[11][11] = P[10][10]; </pre>
17	NavEKF3_core::FuseRngBcn()	range innovations; correct NED position states	NavEKF3_core::SelectRngBcnFusion()	Function call	<pre> void NavEKF3_core::FuseRngBcn() { // declarations ftype pn; ftype pe; ftype pd; ftype bcn_pn; ftype bcn_pe; ftype bcn_pd; </pre>
18	NavEKF3_core::Log_Write_XKF1()	getEulerAngles(); getVelNED(); getLLH(); AP::logger().Write*	NavEKF3_core::Log_Write()	Function call	<pre> void NavEKF3_core::Log_Write_XKF1(uint64_t time_us) const { // Write first EKF packet Vector3f euler; Vector2p posNE; postype_t posD; Vector3f velNED; Vector3f gyroBias; float posDownDeriv; </pre>
19	NavEKF3::getFilterStatus()	core[].getFilterStatus()	Copter::position_ok(); ekf_check()	Function call	<pre> void NavEKF3::getFilterStatus(nav_filter_status &status) const { if (core) { core[primary].getFilterStatus(status); } else { memset(&status, 0, sizeof(status)); } } </pre>
20	NavEKF2_core::UpdateFilter()	Select*Fusion(); CovariancePrediction()	NavEKF2::UpdateFilter() (frontend)	Function call	<pre> void NavEKF2_core::UpdateFilter(bool predict) { // Set the flag to indicate to the filter that the front-end has given permission for a new state prediction cycle to be started startPredictEnabled = predict; // don't run filter updates if states have not been initialised if (!statesInitialised) { return; } </pre>

Ref #	Component/Function	Directly Calls	Directly Called By	Dependency Type	Code Snippet
21	NavEKF2_core::FuseVelPosNED()	Kalman gain on NED pos/vel innovations	NavEKF2_core::SelectVelPosFusion()	Function call	<pre>void NavEKF2_core::FuseVelPosNED() { // health is set bad until test passed bool velHealth = false; // boolean true if velocity measurements have passed innovation consistency check bool posHealth = false; // boolean true if position measurements have passed innovation consistency check bool hgtHealth = false; // boolean true if height measurements have passed innovation consistency check // declare variables used to check measurement errors Vector3F velInnov;</pre>
22	EKFGSF_yaw::update()	predict()/correct() per model; AHRS complementary filter	NavEKF3_core::runYawEstimatorPrediction()	Function call	<pre>void EKFGSF_yaw::update(const Vector3F &delAng, const Vector3F &delVel, const ftype delAngDT, const ftype delVelDT, bool runEKF, ftype TAS) { // copy to class variables</pre>
23	AC_WPNav::set_wp_destination_loc()	get_vector_NEU_cm(); set_wp_destination_NEU_cm()	ArduCopter mode_auto / mode_guided	Function call	<pre>bool AC_WPNav::set_wp_destination_loc(const Location& destination) { bool is_terrain_alt; Vector3f dest_neu_cm; // convert destination location to vector if (!get_vector_NEU_cm(destination, dest_neu_cm, is_terrain_alt)) { return false; }</pre>
24	AP_Mission::get_next_nav_cmd()	get_next_cmd(); read_cmd_from_storage()	ArduCopter/Plane mission sequencing	Function call	<pre>bool AP_Mission::get_next_nav_cmd(uint16_t start_index, Mission_Command& cmd) { // search until the end of the mission command list for (uint16_t cmd_index = start_index; cmd_index < (unsigned)_cmd_total; cmd_index++) { // get next command if (!get_next_cmd(cmd_index, cmd, false)) { // no more commands so return failure return false; } // if found a "navigation" command then return it</pre>
25	AP_L1_Control::update_waypoint()	_ahrs.get_location(_current_loc)	ArduPlane navigation.cpp update loop	Function call	<pre>// Calculate L1 gain required for specified damping float K_L1 = 4.0f * _L1_damping * _L1_damping; // Get current position and velocity if (_ahrs.get_location(_current_loc) == false) { // if no GPS loc available, maintain last nav/target_bearing _data_is_stale = true; return; }</pre>
26	AP_L1_Control::nav_roll_cd()	_ahrs.get_pitch_rad(); _latAccDem	Plane::calc_nav_roll() (Attitude.cpp L608)	Function call	<pre>int32_t AP_L1_Control::nav_roll_cd(void) const { float ret; /* formula can be obtained through equations of balanced spiral: liftForce * cos(roll) = gravityForce * cos(pitch); liftForce * sin(roll) = gravityForce * lateralAcceleration / gravityAcceleration; // as mass = gravityForce/gravityAcceleration see issue 24319 [https://github.com/ArduPilot/ardupilot/issues/24319] Multiplier 100.0f is for converting degrees to centidegrees Made changes to avoid zero division as proposed by Andrew Tridgell: https://github.com/ArduPilot/ardupilot/pull/24331#discussion_r1267798397</pre>

Ref #	Component/Function	Directly Calls	Directly Called By	Dependency Type	Code Snippet
27	AP_TECs::update_pitch_throttle()	<code>_update_pitch(); _update_throttle_with_airspeed()</code>	ArduPlane stabilize/auto pitch loop	Function call	<pre>void AP_TECs::update_pitch_throttle(int32_t hgt_dem_cm, int32_t EAS_dem_cm, enum AP_FixedWing::FlightStage flight_stage, float distance_beyond_land_wp, int32_t ptchMinC0_cd, int16_t throttle_nudge, float hgt_afe, float load_factor, float pitch_trim_deg) {</pre>
28	AC_AttitudeControl::input_euler_angle_roll_pitch_yaw_cd()	<code>input_euler_angle_roll_pitch_yaw_rad()</code>	ArduCopter/ArduPlane flight-mode run()	Function call	<pre>void AC_AttitudeControl::input_euler_angle_roll_pitch_yaw_cd(float euler_roll_angle_cd, float euler_pitch_angle_cd, float euler_yaw_angle_cd, bool slew_yaw) { // Convert from centidegrees on public interface to radians const float euler_roll_angle_rad = cd_to_rad(euler_roll_angle_cd); const float euler_pitch_angle_rad = cd_to_rad(euler_pitch_angle_cd); const float euler_yaw_angle_rad = cd_to_rad(euler_yaw_angle_cd); input_euler_angle_roll_pitch_yaw_rad(euler_roll_angle_rad, euler_pitch_angle_rad, euler_yaw_angle_rad, slew_yaw); }</pre>

Layer 2 — Full transitive chain to final consumer (hop count)

Ref #	PNT Origin	Full Dependency Chain	Final Consumer	Chain Depth	Notes
1	AP_GPS (via AP_AHRS) — user example	AP_GPS AP_AHRS::get_location() Plane::navigate() [ArduPlane/navigation.cpp:86] → → → ModeAuto::navigate() [ArduPlane/mode_auto.cpp:124]	ModeAuto::navigate() [ArduPlane/mode_auto.cpp:124]	3	USER-SUPPLIED example (AAP 0.1.3/0.5.4). The literal example named 'ArduPlane::flight_mode_auto()', which does NOT exist in this snapshot; normalized to the real symbols Plane::navigate() and ModeAuto::navigate(). navigate() is gated on have_position (Table 5 #5).
2	AP_AHRS::groundspeed()	AP_GPS velocity NavEKF3 fusion AP_AHRS::groundspeed() Rover speed control → → → SRV_Channels	SRV_Channels	4	Ground-vehicle speed branch.
3	AP_AHRS::have_inertial_nav()	EKF health AP_AHRS::have_inertial_nav() Copter::ekf_has_absolute_position position_ok() → → → set_mode gate →	mode transition (allow/deny)	4	Also observed in Group 2 (Table 5 #7).
4	AP_AHRS::get_relative_position_NED_origin()	EKF AHRS NED dispatch {AR_PosControl AP_InertialNav} actuators → → →	SRV_Channels / AP_Motors	3	Geodetic↔NED transform branch.
5	AP_AHRS::get_velocity_NED()	EKF AHRS Copter::crash_check / controllers disarm / AP_Motors → → →	AP_Motors / disarm	3	Velocity consumers.
6	NavEKF3_core::UpdateFilter()	{AP_GPS, AP_InertialSensor, AP_Baro} NavEKF3::UpdateFilter AP_AHRS estimate all → → → consumers	AP_Motors	3	[CIRCULAR] AHRS↔EKF; EKF reads AP_DAL.
7	NavEKF3_PosVel fusion (Select)	readGpsData buffer SelectVelPosFusion FuseVelPosNED EKF states AP_AHRS → → → →	AP_AHRS	4	Core fusion scheduling.
8	NavEKF3_core::FuseVelPosNED()	GPS NED innovations FuseVelPosNED stateStruct correction getLLH → → → → AP_AHRS::get_location()	AP_AHRS	4	Position/velocity measurement-update.
9	NavEKF3_core::readGpsData()	AP_GPS::location()/velocity() readGpsData storedGPS buffer SelectVelPosFusion → → → → FuseVelPosNED	EKF state correction	4	Explicit GPS→EKF ingestion hop (cf. Table 1a chains).
10	NavEKF3_core::setAidingMode()	GPS health setAidingMode PV_AidingMode SelectVelPosFusion EKF states → → → →	EKF aiding state	4	Nav aiding-mode control.
11	NavEKF3_core::SelectMagFusion()	magnetometer SelectMagFusion FuseMagnetometer yaw state AP_AHRS heading → → → →	AP_AHRS	4	Heading component.
12	NavEKF3_core::SelectTasFusion()	airspeed SelectTasFusion FuseAirspeed wind/vel states AP_AHRS::wind_estimate → → → →	AP_AHRS	4	Wind/velocity.
13	NavEKF3_core::SelectFlowFusion()	optical flow SelectFlowFusion FuseOptFlow vel/pos states AP_AHRS → → → →	AP_AHRS	4	GPS-denied aiding.
14	NavEKF3_core::getLLH()	EKF state getLLH NavEKF3::getLLH (frontend) AP_AHRS::get_location() nav consumers → → → →	AP_AHRS / consumers	4	EKF position output edge.
15	NavEKF3_core::calcGpsGoodToAlign()	AP_GPS metrics calcGpsGoodToAlign gpsGoodToAlign setAidingMode EKF origin set → → → →	EKF origin / aiding	4	GPS quality gate.
16	NavEKF3_core::resetGyroBias()	InitialiseFilterBootstrap resetGyroBias stateStruct.gyro_bias state prediction → → →	EKF prediction	3	Bias init.
17	NavEKF3_core::FuseRngBcn()	AP_Beacon ranges FuseRngBcn EKF NED position correction AP_AHRS → → →	AP_AHRS	3	ArduPilot_PNT_Reference_Audit.pdf Non-GPS position fusion.

Ref #	PNT Origin	Full Dependency Chain	Final Consumer	Chain Depth	Notes
18	NavEKF3_core::Log_Write_XKF1()	EKF states Log_Write_XKF1 AP_Logger dataflash/SD log → → →	dataflash log	3	Logging Sink branch.
19	NavEKF3::getFilterStatus()	EKF core status bits NavEKF3::getFilterStatus AP_AHRS Copter::position_ok()/ekf_check() → → →	vehicle health logic	3	Status output feeds Group 2.
20	NavEKF2_core::UpdateFilter()	{AP_GPS, INS, Baro} NavEKF2::UpdateFilter AP_AHRS (when EKF2 active) consumers → → →	AP_Motors	3	[CIRCULAR][DUPLICATED] parallel EKF to EKF3.
21	NavEKF2_core::FuseVelPosNED()	GPS NED NavEKF2 FuseVelPosNED EKF2 states AP_AHRS → → →	AP_AHRS	3	[DUPLICATED] mirrors EKF3 #8.
22	EKFGSF_yaw::update()	AP_GPS velocity + IMU EKFGSF_yaw::update emergency yaw EKF yaw reset AP_AHRS → → → →	AP_AHRS	4	Yaw-recovery branch.
23	AC_WPNav	AP_AHRS pos AC_WPNav target AC_PosControl AC_AttitudeControl AP_Motors → → → →	AP_Motors	4	Multirotor waypoint path.
24	AP_Mission	mission storage get_next_nav_cmd mode_auto AC_WPNav controllers → → → →	AP_Motors	4	Mission sequencing path.
25	AP_L1_Control::update_waypoint()	AP_AHRS pos L1 guidance nav_roll_cd() Plane::calc_nav_roll SRV_Channels → → → →	SRV_Channels	4	Fixed-wing lateral path.
26	AP_L1_Control::nav_roll_cd()	L1 latAccDem nav_roll_cd() Plane::calc_nav_roll SRV_Channels → → →	SRV_Channels	3	Roll-demand branch.
27	AP_TECS	airspeed/alt demand TECS update_pitch_throttle pitch/throttle SRV_Channels → → →	SRV_Channels	3	Energy/altitude branch.
28	AC_AttitudeControl (Sink)	nav demand input_euler_* attitude_controller_run_quat rate_controller_run AP_Motors → → → →	AP_Motors	4	ArduPilot_PNT_Reference_Audit.pdf Inbound chain to attitude Sink.

Table 3 — Core Timing (HAL clock primitives, RTC & time-source arbitration, jitter correction, scheduler, GPS time-of-week)

#	File Path	Line(s)	Function/Class	Role	Code Snippet (5-10 lines)	Notes
1	libraries/AP_HAL/system.h	15-20	AP_HAL::micros() / millis() / micros64() / millis64()	Source	<pre>uint16_t micros16(); uint32_t micros(); uint32_t millis(); uint16_t millis16(); uint64_t micros64(); uint64_t millis64();</pre>	Monotonic time backbone (user-cited micros()/millis()). Free functions in the AP_HAL namespace backed by the board RTOS timer; the time primitive every PNT subsystem stamps against.
2	libraries/AP_RTC/AP_RTC.h	32-40	AP_RTC::get_utc_usec()	Source	<pre>/* * get clock in UTC microseconds. Returns false if it is not available. */ bool get_utc_usec(uint64_t &usec) const; // set the system time. If the time has already been set by // something better (according to source_type), this set will be // ignored. void set_utc_usec(uint64_t time_utc_usec, source_type type);</pre>	UTC wall-clock time in microseconds (returns false if unavailable). The absolute-time API consumed for logging and GPS-time conversion.
3	libraries/AP_RTC/AP_RTC.cpp	56-65	AP_RTC::set_utc_usec()	Transform	<pre>// only allow time to be moved forward from the same sourcetype // while the vehicle is disarmed: if (hal.util->get_soft_armed()) { if (type >= rtc_source_type) { // e.g. system-time message when we've been set by the GPS return; } } else { // vehicle is disarmed; accept (e.g.) GPS time source updates if (type > rtc_source_type) {</pre>	Time-SOURCE ARBITRATION: accepts a new UTC time only if its source_type out-ranks the current rtc_source_type (GPS > GCS > HW). Selects the best clock source.
4	libraries/AP_RTC/JitterCorrection.cpp	50-58	JitterCorrection::correct_offboard_timestamp_usec()	Transform	<pre>uint64_t JitterCorrection::correct_offboard_timestamp_usec(uint64_t offboard_usec, uint64_t local_usec) { int64_t diff_us = int64_t(local_usec) - int64_t(offboard_usec); if (!initialised diff_us < link_offset_usec) { // this message arrived from the remote system with a // timestamp that would imply the message was from the // future. We know that isn't possible, so we adjust down the</pre>	Time synchronization: maps an offboard (remote-system) timestamp onto the local clock by tracking the minimum observed link offset — rejects 'from the future' samples.

#	File Path	Line(s)	Function/Class	Role	Code Snippet (5-10 lines)	Notes
5	libraries/AP_Scheduler/AP_Scheduler.cpp	354-363	AP_Scheduler::loop()	Transform	<pre>hal.util->persistent_data.scheduler_task = -1; _loop_sample_time_us = AP_HAL::micros64(); const uint32_t sample_time_us = uint32_t(_loop_sample_time_us); if (_loop_timer_start_us == 0) { _loop_timer_start_us = sample_time_us; _last_loop_time_s = get_loop_period_s(); } else { _last_loop_time_s = (sample_time_us - _loop_timer_start_us) * 1.0e-6;</pre>	Main-loop timing core: stamps each loop with AP_HAL::micros64() and derives the measured loop time. Drives the cadence of every periodic PNT task (EKF predict/fuse, AHRS, control).
6	libraries/AP_Scheduler/AP_Scheduler.cpp	168-176	AP_Scheduler::tick()	Source	<pre>void AP_Scheduler::tick(void) { _tick_counter++; _tick_counter32++; } #if CONFIG_HAL_BOARD == HAL_BOARD_SITL /* fill stack with NaN so we can catch use of uninitialised stack</pre>	ArduPilot_PNT_Reference_Audit.pdf Increments the loop tick counters (_tick_counter / _tick_counter32) that index task scheduling intervals.
7	libraries/AP_Scheduler/AP_Scheduler.h	149-155	AP_Scheduler::get_loop_period_us()	Transform	<pre>// get the time-allowed-per-loop in microseconds uint32_t get_loop_period_us() { if (_loop_period_us == 0) { _loop_period_us = 1000000UL / _loop_rate_hz; } return _loop_period_us; }</pre>	Loop-rate→period derivation: _loop_period_us = 1000000UL / _loop_rate_hz. Couples the configured loop rate to the per-loop time budget and to controller/EKF dt.
8	libraries/AP_GPS/AP_GPS.h	409-418	AP_GPS::time_week_ms() / time_week()	Source	<pre>// GPS time of week in milliseconds uint32_t time_week_ms(uint8_t instance) const { return state[instance].time_week_ms; } uint32_t time_week_ms() const { return time_week_ms(primary_instance); } // GPS week uint16_t time_week(uint8_t instance) const {</pre>	[MULTI-CATEGORY] GPS time-of-week (ms-of-week + week number) — the TIMING facet of the GPS_State that also carries Positioning (Table 1 #4/#17). Emitted separately here per the multi-category rule.
9	libraries/AP_HAL/Scheduler.h	13-21	AP_HAL::Scheduler::delay() / delay_microseconds()	Source	<pre>Scheduler() {} virtual void init() = 0; virtual void delay(uint16_t ms) = 0; /* delay for the given number of microseconds. This needs to be as accurate as possible - preferably within 100 microseconds. */ virtual void delay_microseconds(uint16_t us) = 0;</pre>	Abstract HAL timing-service interface (pure virtual; implemented by the board backend — ChibiOS / Linux / SITL). Supplies the time-blocking primitives delay() / delay_microseconds() and the periodic-task registration register_timer_process() / register_io_process() (lines 60-64) on which the main loop is built. Distinct from the AP_Scheduler library (#5-#7) and the AP_HAL::micros()/millis() clock primitives in system.h (#1).

Table 3a — Timing dependencies (Layer 1 direct edges + Layer 2 transitive chains)

Layer 1 — Direct calls / called-by with typed dependency

Ref #	Component/Function	Directly Calls	Directly Called By	Dependency Type	Code Snippet
1	AP_HAL::micros() / millis()	board RTOS/HAL timer backend	EKF, AP_Scheduler, AP_GPS, all failsafes (ubiquitous)	Function call	<pre>uint16_t micros16(); uint32_t micros(); uint32_t millis(); uint16_t millis16(); uint64_t micros64(); uint64_t millis64();</pre>
2	AP_RTC::get_utc_usec() — via AP::rtc()	AP::rtc().get_utc_usec(rtc)	AP_Scheduler::Log_Write_Performance ; AP_Logger; AP_Stats; AP_NMEA_Output	Shared global / singleton	<pre>/* get clock in UTC microseconds. Returns false if it is not available. */ bool get_utc_usec(uint64_t &usec) const; // set the system time. If the time has already been set by // something better (according to source_type), this set will be // ignored. void set_utc_usec(uint64_t time_utc_usec, source_type type);</pre>

Ref #	Component/Function	Directly Calls	Directly Called By	Dependency Type	Code Snippet
3	AP_RTC::set_utc_usec() (source arbitration)	reads hal.util->get_soft_armed(); compares & writes rtc_source_type	AP_GPS (GPS time); GCS SYSTEM_TIME; AP_BoardConfig (HW RTC)	Data dependency	<pre>// only allow time to be moved forward from the same sourcetype // while the vehicle is disarmed: if (hal.util->get_soft_armed()) { if (type >= rtc_source_type) { // e.g. system-time message when we've been set by the GPS return; } } else { // vehicle is disarmed; accept (e.g.) GPS time source updates if (type > rtc_source_type) {</pre>
4	JitterCorrection::correct_offboard_timestamp_usec()	computes diff_us; updates link_offset_usec	AP_RTC; DroneCAN/MAVLink timestamp consumers	Function call	<pre>uint64_t JitterCorrection::correct_offboard_timestamp_usec(uint64_t offboard_usec, uint64_t local_usec) { int64_t diff_us = int64_t(local_usec) - int64_t(offboard_usec); if (!initialised diff_us < link_offset_usec) { // this message arrived from the remote system with a // timestamp that would imply the message was from the // future. We know that isn't possible, so we adjust down the</pre>
5	AP_Scheduler::loop()	AP_HAL::micros64(); get_loop_period_s(); dispatches run-table tasks	AP_Vehicle::loop() via scheduler.loop() (AP_Vehicle.cpp:545)	Function call	<pre>hal.util->persistent_data.scheduler_task = -1; _loop_sample_time_us = AP_HAL::micros64(); const uint32_t sample_time_us = uint32_t(_loop_sample_time_us); if (_loop_timer_start_us == 0) { _loop_timer_start_us = sample_time_us; _last_loop_time_s = get_loop_period_s(); } else { _last_loop_time_s = (sample_time_us - _loop_timer_start_us) * 1.0e-6;</pre>
6	AP_Scheduler::tick()	increments _tick_counter / _tick_counter32	AP_Scheduler::loop(); fast-loop	Function call	<pre>void AP_Scheduler::tick(void) { _tick_counter++; _tick_counter32++; } #if CONFIG_HAL_BOARD == HAL_BOARD_SITL /* fill stack with NaN so we can catch use of uninitialised stack</pre>
7	AP_Scheduler::get_loop_period_us()	derives _loop_period_us from _loop_rate_hz	task budgeting; controller dt (G_Dt)	Data dependency	<pre>// get the time-allowed-per-loop in microseconds uint32_t get_loop_period_us() { if (_loop_period_us == 0) { _loop_period_us = 1000000UL / _loop_rate_hz; } return _loop_period_us; }</pre>
8	AP_GPS::time_week_ms() / time_week()	returns state[instance].time_week_ms / time_week	NavEKF3 readGpsData (GPS time); time_epoch conversion; logging	Data dependency	<pre>// GPS time of week in milliseconds uint32_t time_week_ms(uint8_t instance) const { return state[instance].time_week_ms; } uint32_t time_week_ms() const { return time_week_ms(primary_instance); } // GPS week uint16_t time_week(uint8_t instance) const {</pre>
9	AP_HAL::Scheduler (abstract HAL timing / scheduling interface)	pure virtual — implemented by the board backend (ChibiOS / Linux / SITL HAL Scheduler)	AP_Scheduler via hal.scheduler->delay_microseconds (AP_Scheduler.cpp:373); register_io_process; AP_Vehicle init; all subsystems	Inheritance / Interface	<pre>Scheduler() {} virtual void init() = 0; virtual void delay(uint16_t ms) = 0; /* delay for the given number of microseconds. This needs to be as accurate as possible - preferably within 100 microseconds. */ virtual void delay_microseconds(uint16_t us) = 0;</pre>

Layer 2 — Full transitive chain to final consumer (hop count)

Ref #	PNT Origin	Full Dependency Chain	Final Consumer	Chain Depth	Notes
1	RTOS/board hardware timer	board timer AP_HAL::micros()/millis() AP_Scheduler / EKF / AP_GPS timestamps every → → → periodic loop	all periodic PNT subsystems	3	ArduPilot_PNT_Reference_Audit.pdf Monotonic-time backbone.
2	GPS/GCS UTC time source	time source AP_RTC AP::rtc().get_utc_usec() AP_Logger / AP_Stats log timestamps → → → →	dataflash log timestamps	4	Singleton consumption edge (AP::rtc()).
3	AP_GPS time / GCS SYSTEM_TIME	time source set_utc_usec source arbitration rtc_source_type system UTC clock → → →	RTC system clock	3	Best-source selection.
4	Offboard (remote) timestamp	MAVLink/DroneCAN timestamp correct_offboard_timestamp_usec corrected local time EKF / → → → sensor fusion	fused-sensor timing	3	Time-sync transform.
5	AP_HAL::micros64() (loop clock)	AP_HAL::micros64() AP_Scheduler::loop() run-table tasks (EKF/AHRS/controllers) → → → AP_Motors	AP_Motors (all tasks)	3	Loop drives every PNT task.
6	AP_Scheduler::loop()	loop() tick() counters task-table interval check periodic task dispatch → → →	periodic task dispatch	3	Cadence counter.
7	SCHED_LOOP_RATE parameter	_loop_rate_hz get_loop_period_us (1e6/rate) task budget + G_Dt AC_AttitudeControl / EKF → → → dt	controller / EKF dt	3	Rate→period coupling.
8	GPS receiver (time-of-week)	GPS receiver GPS_State.time_week AP_GPS::time_week() NavEKF3 readGpsData / → → → time_epoch_usec UTC/log →	EKF GPS-time / UTC conversion	4	[MULTI-CATEGORY] timing facet of GPS.
9	from Group 1), and Behavior board RTOS scheduler (HAL backend)	Change Description. board HAL Scheduler backend AP_HAL::Scheduler interface (delay / register_timer_process) → → AP_Scheduler::loop periodic PNT tasks (EKF / AHRS / control) →	periodic PNT task execution	3	ArduPilot_PNT_Reference_Audit.pdf HAL timing interface beneath the AP_Scheduler library (#5-#7); the board backend implements the pure-virtual interface.

GROUP 2 — INDIRECT / RELATIONAL PNT REFERENCES

INDIRECT / RELATIONAL — not core. This code does not directly read or write PNT data; its behavior changes as a result of PNT state. Each row records Trigger Condition, PNT State Observed (the discriminator from Group 1), and Behavior Change Description.

Table 4 — Indirect / Relational — Positioning (mode position-gating, geofence, GPS-denied fallback)

#	File Path	Line(s)	Function/Class	Trigger Condition	PNT State Observed	Behavior Change Description	Code Snippet (5-10 lines)	Notes
1	ArduCopter/system.cpp	226-234	Copter::position_ok()	Evaluated on position-mode entry and by position-dependent failsafes	failsafe.ekf flag; ekf_has_absolute_position() / ekf_has_relative_position() (EKF position validity)	Returns false when EKF failsafe is set or no EKF position estimate exists, causing callers to deny position-requiring behavior.	<pre>bool Copter::position_ok() const { // return false if ekf failsafe has triggered if (failsafe.ekf) { return false; } // check ekf position estimate return (ekf_has_absolute_position() ekf_has_relative_position()); }</pre>	INDIRECT/RELATIONAL (positioning). Pure observer of EKF position health — reads no raw PNT, but gates downstream behavior.
2	ArduCopter/mode.cpp	303-310	Copter::set_mode() — requires_GPSgate	Pilot/GCS/auto requests a flight-mode change	new_flightmode->requires_GPS() && copter.position_ok()	Rejects the mode change with “requires position” when the target mode needs GPS and position_ok() is false.	<pre>if (!ignore_checks && new_flightmode->requires_GPS() && !copter.position_ok()) { mode_change_failed(new_flightmode, "requires position"); return false; }</pre>	INDIRECT/RELATIONAL (positioning). Central position-gate for ALL position-requiring Copter modes (per-mode requires_GPS() enumerated in the coverage register).

#	File Path	Line(s)	Function/Class	Trigger Condition	PNT State Observed	Behavior Change Description	Code Snippet (5-10 lines)	Notes
3	ArduCopter/fence.cpp	20-28	Copter::fence_checks_async()	25 Hz fence rate-gate elapsed and no breach latched awaiting the main loop	fence.get_breaches() (current breach bitmask, L21); fence.check() (re-evaluates vehicle position vs fences, L25)	Latches new_breaches bitmask for the main loop and logs a breach-cleared event when the vehicle returns inside the fence.	<pre>fence_breaches.last_check_ms = now; const uint8_t orig_breaches = fence.get_breaches(); bool is_landing_or_landed = flightmode->is_landing() ap.land_complete !motors->armed(); // check for new breaches; new_breaches is bitmask of fence types breached fence_breaches.new_breaches = fence.check(is_landing_or_landed); if (!fence_breaches.new_breaches && orig_breaches && fence.get_breaches() == 0) { if (!copter.ap.land_complete) {</pre>	[FIX F1] Cited window centers on get_breaches() (L21) and check() (L25) — the actual geofence observation — not the L8-17 async/timing entry. [DUPLICATED] mirrors Rover::fence_checks_async (#9). Timing rate-gate documented in Table 6.
4	libraries/AC_Fence/AC_Fence.cpp	677-685	AC_Fence::check_fence_polygon()	Polygon (inclusion/exclusion) fence enabled	AP::ahrs().get_location(loc) — current vehicle Location tested against the polygon	record_breach(POLYGON) when _poly_loader.breached(loc) is true sets the polygon breach → bit.	<pre>const bool was_breached = _breached_fences & AC_FENCE_TYPE_POLYGON; Location loc; const bool have_location = AP::ahrs().get_location(loc); if (have_location && _poly_loader.breached(loc, _polygon_breach_distance)) { if (!was_breached) { record_breach(AC_FENCE_TYPE_POLYGON); return true;</pre>	ArduPilot_PNT_Reference_Audit.pdf [FIX F2] Real position-observing path (was mis-cited to L753-762 mask init). Snippet centers on the AP::ahrs().get_location() read (L680).
5	libraries/AC_Fence/AC_Fence.cpp	707-715	AC_Fence::check_fence_circle()	Circle fence enabled	AP::ahrs().get_relative_position_NE_home(home) — NE distance from home (L713)	Records a circle breach when the home distance exceeds the configured radius.	<pre>if (!get_enabled_fences() & AC_FENCE_TYPE_CIRCLE) { // not enabled; no breach return false; } Vector2f home; if (AP::ahrs().get_relative_position_NE_home(home)) { // we (may) remain breached if we can't update home _home_distance = home.length();</pre>	[FIX F2] Real position-observing path; snippet centers on the NE-home position read.
6	libraries/AC_Fence/AC_Fence.cpp	540-548	AC_Fence::check_fence_alt_max()	Maximum-altitude fence enabled	AP::ahrs().get_relative_position_D_home(alt) — altitude relative to home (L546)	Records an altitude breach when alt exceeds _alt_max.	<pre>if (!get_enabled_fences() & AC_FENCE_TYPE_ALT_MAX) { // not enabled; no breach return false; } float alt; AP::ahrs().get_relative_position_D_home(alt); const float _curr_alt = -alt; // translate Down to Up</pre>	[FIX F2] Real altitude-observing path; snippet centers on the D-home altitude read.
7	libraries/AC_Fence/AC_Fence.cpp	837-845	AC_Fence::check()	Periodic fence check invoked from the vehicle fence_checks_async()	(aggregator) results of check_fence_alt_max() / check_fence_circle() / check_fence_polygon() position checks	Returns the bitmask of breached fence types that drives the vehicle fence failsafe action.	<pre>if (!disabled_fences & AC_FENCE_TYPE_CIRCLE) && check_fence_circle()) { ret = AC_FENCE_TYPE_CIRCLE; } // polygon fence check if (!disabled_fences & AC_FENCE_TYPE_POLYGON) && check_fence_polygon()) { ret = AC_FENCE_TYPE_POLYGON; }</pre>	[FIX F2] Orchestrator dispatch (L837 circle, L842 polygon). The position reads themselves are in the dispatched checks (#4-#6).
8	ArduCopter/mode_guided_nogps.cpp	9-15	ModeGuidedNoGPS::init()	GUIDED_NOGPS entry (companion-computer angle control without GPS)	[ABSENT — by design] no position state observed; requires_GPS() returns false (ArduCopter/mode.h)	Starts angle control only and remains available when position_ok() is false — the GPS-denied fallback path.	<pre>// initialise guided_nogps controller bool ModeGuidedNoGPS::init(bool ignore_checks) { // start in angle control mode ModeGuided::angle_control_start(); return true; }</pre>	INDIRECT/RELATIONAL (positioning). Contrast row: intentionally PNT-position-INDEPENDENT mode (documents an explicit absence of positioning dependency).

#	File Path	Line(s)	Function/Class	Trigger Condition	PNT State Observed	Behavior Change Description	Code Snippet (5-10 lines)	Notes
9	Rover/fence.cpp	17-25	Rover::fence_checks_async()	Rover fence rate-gate elapsed and no breach latched	fence.get_breaches() (L19); fence.check() (vehicle position vs fence, L21)	Latches new_breaches for the main loop; clears on return inside the fence.	<pre>fence_breaches.last_check_ms = now; const uint8_t orig_breaches = fence.get_breaches(); // check for new breaches; new_breaches is bitmask of fence types breached fence_breaches.new_breaches = fence.check(); if (!fence_breaches.new_breaches && orig_breaches && fence.get_breaches() == 0) { // record clearing of breach LOGGER_WRITE_ERROR(LogErrorSubsystem::FAILS AFE_FENCE, LogErrorCode::ERROR_RESOLVED); }</pre>	ArduPilot_PNT_Reference_Audit.pdf [DUPLICATED] structurally identical to ArduCopter::fence_checks_async (#3) — copy-paste across vehicles (extraction risk).
10	ArduPlane/takeoff.cpp	58-62	Plane::auto_takeoff_check()	Auto/TAKEOFF launch readiness re-evaluated each loop from Plane::set_servos() (ArduPlane/servos.cpp:125)	gps.status() < AP_GPS::GPS_OK_FIX_3D (GPS fix quality; gps.status() = AP_GPS::status())	Returns false — auto-takeoff is blocked (throttle stays suppressed) until a 3D GPS fix is acquired.	<pre>// Check for bad GPS if (gps.status() < AP_GPS::GPS_OK_FIX_3D) { // no auto takeoff without GPS lock return false; }</pre>	INDIRECT/RELATIONAL (positioning). User-example PNT-State-Observed accessor gps.status() (= AP_GPS::status(), libraries/AP_GPS/AP_GPS.h:293). [DUPLICATED] the same gps.status() < GPS_OK_FIX_3D fix-gate recurs in ArduPlane/Plane.cpp (L420/485) and ArduPlane/is_flying.cpp (extraction risk).

Table 4a — Indirect Positioning dependencies (Layer 1 + Layer 2)

Layer 1 — Direct calls / called-by with typed dependency

Ref #	Component/Function	Directly Calls	Directly Called By	Dependency Type	Code Snippet
1	Copter::position_ok()	ekf_has_absolute_position(); ekf_has_relative_position(); reads failsafe.ekf	Copter::set_mode (mode.cpp:306); position-dependent failsafes	Function call	<pre>bool Copter::position_ok() const { // return false if ekf failsafe has triggered if (failsafe.ekf) { return false; } // check ekf position estimate return (ekf_has_absolute_position() ekf_has_relative_position()); }</pre>
2	Copter::set_mode() requires_GPS gate	new_flightmode->requires_GPS(); copter.position_ok()	GCS/RC/auto mode-change entry points	Function call	<pre>if (!ignore_checks && new_flightmode->requires_GPS() && !copter.position_ok()) { mode_change_failed(new_flightmode, "requires position"); return false; }</pre>
3	Copter::fence_checks_async()	fence.get_breaches(); fence.check()	Copter IO thread (1 kHz async, 25 Hz gated)	Function call	<pre>fence_breaches.last_check_ms = now; const uint8_t orig_breaches = fence.get_breaches(); bool is_landing_or_landed = flightmode->is_landing() ap.land_complete !motors->armed(); // check for new breaches; new_breaches is bitmask of fence types breached fence_breaches.new_breaches = fence.check(is_landing_or_landed); if (!fence_breaches.new_breaches && orig_breaches && fence.get_breaches() == 0) { if (!copter.ap.land_complete) {</pre>
4	AC_Fence::check_fence_polygon()	AP::ahrs().get_location(); _poly_loader.breached()	AC_Fence::check() (L842)	Shared global / singleton	<pre>const bool was_breached = _breached_fences & AC_FENCE_TYPE_POLYGON; Location loc; const bool have_location = AP::ahrs().get_location(loc); if (have_location && _poly_loader.breached(loc, _polygon_breach_distance)) { if (!was_breached) { record_breach(AC_FENCE_TYPE_POLYGON); return true; }</pre>

Ref #	Component/Function	Directly Calls	Directly Called By	Dependency Type	Code Snippet
5	AC_Fence::check_fence_circle()	AP::ahrs().get_relative_position_NE_home()	AC_Fence::check() (L837)	Shared global / singleton	<pre>if (!(get_enabled_fences() & AC_FENCE_TYPE_CIRCLE)) { // not enabled; no breach return false; } Vector2f home; if (AP::ahrs().get_relative_position_NE_home(home)) { // we (may) remain breached if we can't update home _home_distance = home.length(); }</pre>
6	AC_Fence::check_fence_alt_max()	AP::ahrs().get_relative_position_D_home()	AC_Fence::check() (L821)	Shared global / singleton	<pre>if (!(get_enabled_fences() & AC_FENCE_TYPE_ALT_MAX)) { // not enabled; no breach return false; } float alt; AP::ahrs().get_relative_position_D_home(alt); const float _curr_alt = -alt; // translate Down to Up</pre>
7	AC_Fence::check()	check_fence_alt_max(); check_fence_circle(); check_fence_polygon()	Copter::fence_checks_async(); Rover::fence_checks_async()	Function call	<pre>if (!(disabled_fences & AC_FENCE_TYPE_CIRCLE) && check_fence_circle()) { ret = AC_FENCE_TYPE_CIRCLE; } // polygon fence check if (!(disabled_fences & AC_FENCE_TYPE_POLYGON) && check_fence_polygon()) { ret = AC_FENCE_TYPE_POLYGON; }</pre>
8	ModeGuidedNoGPS::init()	ModeGuided::angle_control_start()	Copter::set_mode()	Inheritance / Interface	<pre>// initialise guided_nogps controller bool ModeGuidedNoGPS::init(bool ignore_checks) { // start in angle control mode ModeGuided::angle_control_start(); return true; }</pre>
9	Rover::fence_checks_async()	fence.get_breaches(); fence.check()	Rover IO thread	Function call	<pre>fence_breaches.last_check_ms = now; const uint8_t orig_breaches = fence.get_breaches(); // check for new breaches; new_breaches is bitmask of fence types breached fence_breaches.new_breaches = fence.check(); if (!fence_breaches.new_breaches && orig_breaches && fence.get_breaches() == 0) { // record clearing of breach LOGGER_WRITE_ERROR(LogErrorSubsystem::FAILSAFE_FENCE, LogErrorCode::ERROR_RESOLVED); }</pre>
10	Plane::auto_takeoff_check()	AP_GPS::status() (gps.status()); compares vs AP_GPS::GPS_OK_FIX_3D	Plane::set_servos() (ArduPlane/servos.cpp:125)	Function call	<pre>// Check for bad GPS if (gps.status() < AP_GPS::GPS_OK_FIX_3D) { // no auto takeoff without GPS lock return false; }</pre>

Layer 2 – Full transitive chain to final consumer (hop count)

Ref #	PNT Origin	Full Dependency Chain	Final Consumer	Chain Depth	Notes
1	EKF position health (via AP_AHRS)	AP_GPS NavEKF3 AP_AHRS::have_inertial_nav() Copter::ekf_has_absolute_position() → → → Copter::position_ok() set_mode gate →	flight-mode entry allow/deny	5	Core positioning-health gate.
2	Copter::position_ok()	position_ok() set_mode requires_GPS gate mode_change_failed() mode unchanged → → →	rejected mode change	3	Gate decision path.
3	AP_AHRS vehicle position (via AC_Fence)	AP::ahrs() position AC_Fence::check (polygon/circle/alt) get_breaches bitmask → → → fence_checks_async latch Copter::fence_check() action →	Copter fence failsafe (RTL/LAND)	4	ArduPilot_PNT_Reference_Audit.pdf [DUPLICATED] mirrors Rover chain #9.
4	AP_AHRS::get_location()	AP_AHRS::get_location() check_fence_polygon_poly_loader.breached() record_breach() → → → → _breached_fences	polygon breach bit	4	Polygon position test.
5	AP_AHRS NE-home position	AP::ahrs().get_relative_position_NE_home() check_fence_circle distance>radius → → → record_breach()	circle breach bit	3	Circle position test.

Ref #	PNT Origin	Full Dependency Chain	Final Consumer	Chain Depth	Notes
6	AP_AHRS D-home altitude	AP::ahrs().get_relative_position_D_home() check_fence_alt_max alt>_alt_max → → → record_breach()	altitude breach bit	3	Altitude position test.
7	AC_Fence position checks	position checks AC_Fence::check() aggregate breach bitmask vehicle fence_check() → → → → failsafe action	vehicle fence failsafe	4	Aggregator path.
8	(no PNT — GPS-independent)	ModeGuidedNoGPS::init() angle_control_start() AC_AttitudeControl AP_Motors → → →	AP_Motors	3	[ABSENT] no positioning hop — fallback by design.
9	AP_AHRS vehicle position (via AC_Fence)	AP::ahrs() position AC_Fence::check get_breaches Rover::fence_checks_async latch → → → → Rover::fence_check() action	Rover fence failsafe	4	[DUPLICATED] structurally identical to Copter chain #3.
10	AP_GPS::status() (GPS fix-status accessor)	AP_GPS::status() Plane::auto_takeoff_check() [ArduPlane/takeoff.cpp:59] Plane::set_servos() → → [ArduPlane/servos.cpp:125]	Plane::set_servos() (throttle suppression / launch gate)	2	ArduPilot_PNT_Reference_Audit.pdf User-example gps.status() observation gating auto-takeoff.

Table 5 — Indirect / Relational — Navigation (EKF-health watchdog, nav-output reporting & consumption, crash/flight detection)

#	File Path	Line(s)	Function/Class	Trigger Condition	PNT State Observed	Behavior Change Description	Code Snippet (5-10 lines)	Notes
1	ArduCopter/ekf_check.cpp	117-125	Copter::ekf_over_threshold()	10 Hz EKF health check while the EKF has an origin	ahrs.get_variances() → velocity / position / height / magnetometer variances vs g.fs_ekf_thresh	Counts bad iterations; after EKF_CHECK_ITERATIONS_MAX (≈10) trips the EKF failsafe (LAND / AltHold).	<pre>float position_var, vel_var, height_var, tas_variance; Vector3f mag_variance; variances_valid = ahrs.get_variances(vel_var, position_var, height_var, mag_variance, tas_variance); if (!variances_valid) { return false; } uint32_t now_us = AP_HAL::micros();</pre>	INDIRECT/RELATIONAL (navigation). [DUPLICATED] EKF-variance watchdog replicated across ArduCopter/ekf_check.cpp, ArduPlane/ekf_check.cpp, Rover/ekf_check.cpp and ArduSub/failsafe.cpp — a 4-way copy-paste (extraction risk).
2	ArduPlane/ekf_check.cpp	56-65	Plane::ekf_check()	10 Hz EKF health check (fixed-wing)	ekf_over_threshold() (compass & velocity variance vs fs_ekf_thresh)	Increments fail_count; two iterations before failsafe asks the EKF to reset yaw (ahrs.request_yaw_reset()); trips EKF failsafe at MAX.	<pre>// compare compass and velocity variance vs threshold if (ekf_over_threshold()) { // if compass is not yet flagged as bad if (!ekf_check_state.bad_variance) { // increase counter ekf_check_state.fail_count++; if (ekf_check_state.fail_count == (EKF_CHECK_ITERATIONS_MAX-2)) { // we are two iterations away from declaring an EKF failsafe, ask the EKF if we can reset // yaw to resolve the issue ahrs.request_yaw_reset();</pre>	INDIRECT/RELATIONAL (navigation). [DUPLICATED] EKF-variance watchdog replicated across ArduCopter/ekf_check.cpp, ArduPlane/ekf_check.cpp, Rover/ekf_check.cpp and ArduSub/failsafe.cpp — a 4-way copy-paste (extraction risk).
3	Rover/ekf_check.cpp	93-102	Rover::ekf_over_threshold()	10 Hz EKF health check (ground vehicle)	ahrs.get_variances() → compass / velocity / position variance; ≥2 over threshold	Sets bad_variance Rover EKF → failsafe action.	<pre>// use EKF to get variance float position_variance, vel_variance, height_variance, tas_variance; Vector3f mag_variance; ahrs.get_variances(vel_variance, position_variance, height_variance, mag_variance, tas_variance); // return true if two of compass, velocity and position variances are over the threshold uint8_t over_thresh_count = 0; if (mag_variance.length() >= g.fs_ekf_thresh) { over_thresh_count++;</pre>	INDIRECT/RELATIONAL (navigation). [DUPLICATED] EKF-variance watchdog replicated across ArduCopter/ekf_check.cpp, ArduPlane/ekf_check.cpp, Rover/ekf_check.cpp and ArduSub/failsafe.cpp — a 4-way copy-paste (extraction risk).

#	File Path	Line(s)	Function/Class	Trigger Condition	PNT State Observed	Behavior Change Description	Code Snippet (5-10 lines)	Notes
4	ArduSub/failsafe.cpp	112-121	Sub::failsafe_ekf_check()	Periodic EKF health check (submarine)	ahrs.get_variances() → compass_variance & vel_variance vs g.fs_ekf_thresh	Clears/sets failsafe.ekf and AP_Notify ekf_bad based on variance vs threshold; GCS warns “EKF bad”.	<pre> Vector3f magVar; float compass_variance; float vel_variance; ahrs.get_variances(vel_variance, posVar, hgtVar, magVar, tasVar); compass_variance = magVar.length(); if (compass_variance < g.fs_ekf_thresh && vel_variance < g.fs_ekf_thresh) { last_ekf_good_ms = AP_HAL::millis(); failsafe.ekf = false; AP_Notify::flags.ekf_bad = false; </pre>	ArduPilot_PNT_Reference_Audit.pdf INDIRECT/RELATIONAL (navigation). [FIX F13 dep-type] [DUPLICATED] EKF-variance watchdog replicated across ArduCopter/ekf_check.cpp, ArduPlane/ekf_check.cpp, Rover/ekf_check.cpp and ArduSub/failsafe.cpp — a 4-way copy-paste (extraction risk).
5	ArduCopter/crash_check.cpp	65-74	Copter::crash_check() — attitude test	Disarm/crash monitor running while armed and not landed	ahrs.cos_roll() / ahrs.cos_pitch() (lean angle); attitude_control->get_att_error_angle_deg()	Resets crash_counter (declares not-crashing) when lean angle ≤ 15° or attitude error 30°. ≤	<pre> // check for lean angle over 15 degrees const float lean_angle_deg = degrees(acosf(ahrs.cos_roll()*ahrs.cos_pitch())); if (lean_angle_deg <= CRASH_CHECK_ANGLE_MIN_DEG) { crash_counter = 0; return; // check for angle error over 30 degrees const float angle_error = attitude_control->get_att_error_angle_deg(); if (angle_error <= CRASH_CHECK_ANGLE_DEVIATION_DEG) { </pre>	[FIX F3] Split row: attitude-only observation (lean L66, angle-error L73). Velocity observation is the separate row #6 (was wrongly bundled into the L64-73 citation).
6	ArduCopter/crash_check.cpp	78-84	Copter::crash_check() — velocity test	Continues after the attitude test indicates a possible crash	ahrs.get_velocity_NED(vel_ned) NED speed → magnitude (L80-81)	Resets crash_counter when speed ≥ CRASH_CHECK_SPEED_MAX (10 m/s); otherwise increments toward the 2-second crash trigger disarm. →	<pre> // check for speed under 10m/s (if available) Vector3f vel_ned; if (ahrs.get_velocity_NED(vel_ned) && (vel_ned.length() >= CRASH_CHECK_SPEED_MAX)) { crash_counter = 0; return; } </pre>	[FIX F3] Split row: the velocity observation cited at L78-84 (ahrs.get_velocity_NED L80-81), now accurately centered — not the L64-73 attitude range.
7	ArduPlane/GCS_MAVLink_Plane.cpp	204-212	GCS_MAVLINK_Plane::send_nav_controller_output()	GCS telemetry stream requests NAV_CONTROLLER_OUTPUT	nav_controller navigation state (target bearing, xtrack error, wp distance) — unless in MANUAL	Emits the NAV_CONTROLLER_OUTPUT MAVLink message (navigation-health reporting to the GCS).	<pre> void GCS_MAVLINK_Plane::send_nav_controller_output() const { if (plane.control_mode == &plane.mode_manual) { return; } } #if HAL_QUADPLANE_ENABLED const QuadPlane &quadplane = plane.quadplane; if (quadplane.show_vtol_view() && quadplane.using_wp_nav()) { const Vector3f &targets = quadplane.attitude_control->get_att_target_euler_cd(); </pre>	INDIRECT/RELATIONAL (navigation). Reporting Sink for navigation state.
8	ArduPlane/Attitude.cpp	604-611	Plane::calc_nav_roll()	Fixed-wing attitude loop computing the roll demand	nav_controller->nav_roll_cd() (L1/navigation lateral roll demand)	Sets nav_roll_cd from the navigation controller, constrained to ±roll_limit_cd.	<pre> calculate a new nav_roll_cd from the navigation controller */ void Plane::calc_nav_roll() { int32_t commanded_roll = nav_controller->nav_roll_cd(); nav_roll_cd = constrain_int32(commanded_roll, -roll_limit_cd, roll_limit_cd); update_load_factor(); } </pre>	INDIRECT/RELATIONAL (navigation). Consumes navigation-controller output (user-cited calc_nav_roll).
9	ArduPlane/navigation.cpp	86-94	Plane::navigate()	10 Hz fixed-wing navigation update	have_position flag (EKF/GPS position validity); next_WP_loc	Returns early (skips navigation) when have_position is false; otherwise computes the desired direction and distance to the next waypoint.	<pre> void Plane::navigate() { // do not navigate with corrupt data // ----- if (!have_position) { return; } if (next_WP_loc.lat == 0 && next_WP_loc.lng == 0) { </pre>	ArduPilot_PNT_Reference_Audit.pdf INDIRECT/RELATIONAL (navigation). This is the real symbol behind the user-example 'flight_mode_auto' chain (Table 2a L2 #1); gated on PNT position validity.

#	File Path	Line(s)	Function/Class	Trigger Condition	PNT State Observed	Behavior Change Description	Code Snippet (5-10 lines)	Notes
10	ArduPlane/navigation.cpp	304-311	Plane::calc_gndspeed_undershoot()	10 Hz GPS/navigation update (Plane::update_GPS_10Hz, ArduPlane/Plane.cpp:490)	ahrs.have_inertial_nav() (inertial-navigation solution availability)	Ground-speed-undershoot (min-airspeed wind/flyaway compensation) is computed only when inertial nav is available; otherwise it is skipped and left stale/zero.	<pre> Vector3f velNED; if (ahrs.have_inertial_nav() && ahrs.get_velocity_NED(velNED)) { const Matrix3f &rotMat = ahrs.get_rotation_body_to_ned(); Vector2f yawVect = Vector2f(rotMat.a.x,rotMat.b.x); if (!yawVect.is_zero()) { yawVect.normalize(); float gndSpdFwd = yawVect * velNED.xy(); groundspeed_undershoot_is_valid = aparm.min_airspeed > 0; </pre>	INDIRECT/RELATIONAL (navigation). User-example PNT-State-Observed accessor ahrs.have_inertial_nav() (libraries/AP_AHRS/AP_AHRS.h:273); the accessor itself is catalogued in Table 2 #3.

Table 5a — Indirect Navigation dependencies (Layer 1 + Layer 2)**Layer 1 — Direct calls / called-by with typed dependency**

Ref #	Component/Function	Directly Calls	Directly Called By	Dependency Type	Code Snippet
1	Copter::ekf_over_threshold()	ahrs.get_variances(); variance filters; compare g.fs_ekf_thresh	Copter::ekf_check()	Function call	<pre> float position_var, vel_var, height_var, tas_variance; Vector3f mag_variance; variances_valid = ahrs.get_variances(vel_var, position_var, height_var, mag_variance, tas_variance); if (!variances_valid) { return false; } uint32_t now_us = AP_HAL::micros(); </pre>
2	Plane::ekf_check()	ekf_over_threshold(); ahrs.request_yaw_reset()	Plane 10 Hz scheduler	Function call	<pre> // compare compass and velocity variance vs threshold if (ekf_over_threshold()) { // if compass is not yet flagged as bad if (!ekf_check_state.bad_variance) { // increase counter ekf_check_state.fail_count++; if (ekf_check_state.fail_count == (EKF_CHECK_ITERATIONS_MAX-2)) { // we are two iterations away from declaring an EKF failsafe, ask the EKF if we can reset // yaw to resolve the issue ahrs.request_yaw_reset(); </pre>
3	Rover::ekf_over_threshold()	ahrs.get_variances(); compare g.fs_ekf_thresh	Rover::ekf_check()	Function call	<pre> // use EKF to get variance float position_variance, vel_variance, height_variance, tas_variance; Vector3f mag_variance; ahrs.get_variances(vel_variance, position_variance, height_variance, mag_variance, tas_variance); // return true if two of compass, velocity and position variances are over the threshold uint8_t over_thresh_count = 0; if (mag_variance.length() >= g.fs_ekf_thresh) { over_thresh_count++; </pre>
4	Sub::failsafe_ekf_check()	ahrs.get_variances(); AP_HAL::millis(); gcs().send_text()	Sub periodic scheduler	Function call	<pre> Vector3f magVar; float compass_variance; float vel_variance; ahrs.get_variances(vel_variance, posVar, hgtVar, magVar, tasVar); compass_variance = magVar.length(); if (compass_variance < g.fs_ekf_thresh && vel_variance < g.fs_ekf_thresh) { last_ekf_good_ms = AP_HAL::millis(); failsafe.ekf = false; AP_Notify::flags.ekf_bad = false; </pre>

Ref #	Component/Function	Directly Calls	Directly Called By	Dependency Type	Code Snippet
5	Copter::crash_check() (attitude)	ahrs.cos_roll(); ahrs.cos_pitch(); attitude_control->get_att_error_angle_deg()	Copter::crash_check()	Function call	<pre>// check for lean angle over 15 degrees const float lean_angle_deg = degrees(acosf(ahrs.cos_roll()*ahrs.cos_pitch())); if (lean_angle_deg <= CRASH_CHECK_ANGLE_MIN_DEG) { crash_counter = 0; return; } // check for angle error over 30 degrees const float angle_error = attitude_control->get_att_error_angle_deg(); if (angle_error <= CRASH_CHECK_ANGLE_DEVIATION_DEG) {</pre>
6	Copter::crash_check() (velocity)	ahrs.get_velocity_NED(vel_ned)	Copter::crash_check()	Function call	<pre>// check for speed under 10m/s (if available) Vector3f vel_ned; if (ahrs.get_velocity_NED(vel_ned) && (vel_ned.length() >= CRASH_CHECK_SPEED_MAX)) { crash_counter = 0; return; }</pre>
7	GCS_MAVLINK_Plane::send_nav_controller_output()	reads plane.control_mode; nav_controller state	GCS telemetry scheduler	Function call	<pre>void GCS_MAVLINK_Plane::send_nav_controller_output() const { if (plane.control_mode == &plane.mode_manual) { return; } } #ifdef HAL_QUADPLANE_ENABLED const QuadPlane &quadplane = plane.quadplane; if (quadplane.show_vtol_view() && quadplane.using_wp_nav()) { const Vector3f &targets = quadplane.attitude_control->get_att_target_euler_cd();</pre>
8	Plane::calc_nav_roll()	nav_controller->nav_roll_cd(); update_load_factor()	Plane fast attitude loop	Function call	<pre>calculate a new nav_roll_cd from the navigation controller */ void Plane::calc_nav_roll() { int32_t commanded_roll = nav_controller->nav_roll_cd(); nav_roll_cd = constrain_int32(commanded_roll, -roll_limit_cd, roll_limit_cd); update_load_factor(); }</pre>
9	Plane::navigate()	reads have_position; next_WP_loc; nav_controller update	Plane 10 Hz nav loop	Data dependency	<pre>void Plane::navigate() { // do not navigate with corrupt data // ----- if (!have_position) { return; } if (next_WP_loc.lat == 0 && next_WP_loc.lng == 0) {</pre>
10	Plane::calc_gndspeed_undershoot ()	ahrs.have_inertial_nav(); ahrs.get_velocity_NED()	Plane::update_GPS_10Hz() (ArduPlane/Plane.cpp:490)	Function call	<pre>Vector3f velNED; if (ahrs.have_inertial_nav() && ahrs.get_velocity_NED(velNED)) { const Matrix3f &rotMat = ahrs.get_rotation_body_to_ned(); Vector2f yawVect = Vector2f(rotMat.a.x,rotMat.b.x); if (!yawVect.is_zero()) { yawVect.normalize(); float gndSpdFwd = yawVect * velNED.xy(); groundspeed_undershoot_is_valid = aparm.min_groundspeed > 0;</pre>

Layer 2 — Full transitive chain to final consumer (hop count)

Ref #	PNT Origin	Full Dependency Chain	Final Consumer	Chain Depth	Notes
1	EKF variances (via AP_AHRS)	AP_GPS/EKF ahrs.get_variances() ekf_over_threshold() Copter::ekf_check fail_count → → → → failsafe_ekf_event() LAND/AltHold →	EKF failsafe action (LAND)	5	[DUPLICATED] EKF-variance watchdog replicated across ArduCopter/ekf_check.cpp, ArduPlane/ekf_check.cpp, Rover/ekf_check.cpp and ArduSub/failsafe.cpp — a 4-way copy-paste (extraction risk).
2	EKF variances (Plane)	EKF variances Plane::ekf_check bad_variance EKF failsafe mode change (RTL/FBWA) → → → →	Plane EKF failsafe	4	[DUPLICATED] mirrors #1.
3	EKF variances (Rover)	EKF variances Rover::ekf_over_threshold Rover::ekf_check EKF failsafe action → → →	Rover EKF failsafe	3	[DUPLICATED] mirrors #1.

Ref #	PNT Origin	Full Dependency Chain	Final Consumer	Chain Depth	Notes
4	EKF variances (Sub)	EKF variances Sub::failsafe_ekf_check failsafe.ekf flag GCS warn → → →	Sub EKF failsafe flag	3	[DUPLICATED] mirrors #1.
5	AP_AHRS attitude	AHRS attitude crash_check attitude test crash_counter arming.disarm() → → →	motor disarm	3	Attitude branch.
6	AP_AHRS NED velocity	EKF velocity ahrs.get_velocity_NED() crash_check speed test crash_counter → → → → arming.disarm()	motor disarm	4	Velocity branch (F3 split).
7	Navigation controller state	EKF/nav state nav_controller send_nav_controller_output MAVLink GCS → → → →	GCS NAV_CONTROLLER_OUTPUT	4	Reporting branch.
8	Navigation controller roll demand	nav_controller (L1/EKF) nav_roll_cd() calc_nav_roll nav_roll_cd attitude controller → → → → → SRV_Channels	SRV_Channels	5	Roll-demand consumption.
9	EKF position validity	EKF position have_position Plane::navigate nav_controller->update_waypoint → → → → calc_nav_roll SRV_Channels →	SRV_Channels	5	Real symbol behind user-example chain (Table 2a L2 #1).
10	AP_AHRS::have_inertial_nav() (inertial-nav availability)	AP_AHRS::have_inertial_nav() Plane::calc_gndspeed_undershoot() → [ArduPlane/navigation.cpp:305] Plane::update_GPS_10Hz() [ArduPlane/Plane.cpp:490] →	Plane::update_GPS_10Hz() (groundspeed-undershoot nav compensation)	2	ArduPilot_PNT_Reference_Audit.pdf User-example ahrs.have_inertial_nav() observation gating nav groundspeed compensation.

Table 6 — Indirect / Relational — Timing (main-loop-lockup watchdogs, PNT-health message hysteresis, rate gating)

#	File Path	Line(s)	Function/Class	Trigger Condition	PNT State Observed	Behavior Change Description	Code Snippet (5-10 lines)	Notes
1	ArduCopter/failsafe.cpp	37-45	Copter::failsafe_check()	1 kHz timer-failsafe ISR; scheduler tick counter has not advanced since the last call	AP_HAL::micros() (tnow) and AP_Scheduler::ticks() loop counter (timing / loop-cadence state)	When ticks stall and tnow-failsafe_last_timestamp > 2,000,000 us (2 s) sets in_failsafe, drives motors->output_min(), then disarms every 1 s (CPU failsafe).	uint32_t tnow = AP_HAL::micros(); const uint16_t ticks = scheduler.ticks(); if (ticks != failsafe_last_ticks) { // the main loop is running, all is OK failsafe_last_ticks = ticks; failsafe_last_timestamp = tnow; if (in_failsafe) { in_failsafe = false; }	INDIRECT/RELATIONAL (timing). [DUPLICATED] Main-loop-lockup watchdog replicated 5 ways — ArduCopter/failsafe.cpp, ArduPlane/failsafe.cpp, Rover/failsafe.cpp, ArduSub/failsafe.cpp (mainloop_failsafe_check) and Blimp/failsafe.cpp — each reads AP_Scheduler::ticks() + AP_HAL::micros(); divergent thresholds/actions (extraction risk).
2	ArduPlane/failsafe.cpp	22-30	Plane::failsafe_check()	1 kHz timer-failsafe ISR; scheduler tick counter unchanged (main loop stalled)	micros() (tnow) and AP_Scheduler::ticks() (ticks vs last_ticks)	After > 200,000 us (0.2 s) stall sets in_failsafe and passes RC inputs straight to servo outputs (SRV_Channels) every 20 ms instead of disarming.	uint32_t tnow = micros(); const uint16_t ticks = scheduler.ticks(); if (ticks != last_ticks) { // the main loop is running, all is OK last_ticks = ticks; last_timestamp = tnow; in_failsafe = false; return; }	INDIRECT/RELATIONAL (timing). [DUPLICATED] Main-loop-lockup watchdog replicated 5 ways — ArduCopter/failsafe.cpp, ArduPlane/failsafe.cpp, Rover/failsafe.cpp, ArduSub/failsafe.cpp (mainloop_failsafe_check) and Blimp/failsafe.cpp — each reads AP_Scheduler::ticks() + AP_HAL::micros(); divergent thresholds/actions (extraction risk). Divergent: 0.2 s trip + RC-passthrough (vs Copter 2 s + disarm).
3	Rover/failsafe.cpp	22-30	Rover::failsafe_check()	1 kHz timer-failsafe ISR; scheduler tick counter unchanged	AP_HAL::micros() (tnow) and AP_Scheduler::ticks() (ticks vs last_ticks)	After > 200,000 us (0.2 s) stall calls arming.disarm(AP_Arming::Method::CPUFAILSAFE).	const uint32_t tnow = AP_HAL::micros(); const uint16_t ticks = scheduler.ticks(); if (ticks != last_ticks) { // the main loop is running, all is OK last_ticks = ticks; last_timestamp = tnow; return; } }	INDIRECT/RELATIONAL (timing). [DUPLICATED] Main-loop-lockup watchdog replicated 5 ways — ArduCopter/failsafe.cpp, ArduPlane/failsafe.cpp, Rover/failsafe.cpp, ArduSub/failsafe.cpp (mainloop_failsafe_check) and Blimp/failsafe.cpp — each reads AP_Scheduler::ticks() + AP_HAL::micros(); divergent thresholds/actions (extraction risk).

#	File Path	Line(s)	Function/Class	Trigger Condition	PNT State Observed	Behavior Change Description	Code Snippet (5-10 lines)	Notes
4	ArduSub/failsafe.cpp	31-39	Sub::mainloop_failsafe_check()	1 kHz timer-failsafe ISR; scheduler tick counter unchanged	AP_HAL::micros() (tnow) and AP_Scheduler::ticks() loop counter	After > 2,000,000 us (2 s) stall sets in_failsafe and drives motors.output_min().	<pre>uint32_t tnow = AP_HAL::micros(); const uint16_t ticks = scheduler.ticks(); if (ticks != failsafe_last_ticks) { // the main loop is running, all is OK failsafe_last_ticks = ticks; failsafe_last_timestamp = tnow; if (in_failsafe) { in_failsafe = false; } }</pre>	ArduPilot_PNT_Reference_Audit.pdf INDIRECT/RELATIONAL (timing). [DUPLICATED] Main-loop-lockup watchdog replicated 5 ways — ArduCopter/failsafe.cpp, ArduPlane/failsafe.cpp, Rover/failsafe.cpp, ArduSub/failsafe.cpp (mainloop_failsafe_check) and Blimp/failsafe.cpp — each reads AP_Scheduler::ticks() + AP_HAL::micros(); divergent thresholds/actions (extraction risk).
5	Blimp/failsafe.cpp	37-45	Blimp::failsafe_check()	1 kHz timer-failsafe ISR; scheduler tick counter unchanged	AP_HAL::micros() (tnow) and AP_Scheduler::ticks() loop counter	After > 2,000,000 us (2 s) stall sets in_failsafe and drives motors->output_min().	<pre>uint32_t tnow = AP_HAL::micros(); const uint16_t ticks = scheduler.ticks(); if (ticks != failsafe_last_ticks) { // the main loop is running, all is OK failsafe_last_ticks = ticks; failsafe_last_timestamp = tnow; if (in_failsafe) { in_failsafe = false; } }</pre>	INDIRECT/RELATIONAL (timing). [DUPLICATED] Main-loop-lockup watchdog replicated 5 ways — ArduCopter/failsafe.cpp, ArduPlane/failsafe.cpp, Rover/failsafe.cpp, ArduSub/failsafe.cpp (mainloop_failsafe_check) and Blimp/failsafe.cpp — each reads AP_Scheduler::ticks() + AP_HAL::micros(); divergent thresholds/actions (extraction risk). Blimp included only because it observes scheduler timing state (AAP 0.3.2).
6	ArduCopter/ekf_check.cpp	84-88	Copter::ekf_check() — GCS warnthrottle	EKF variance already flagged bad AND 30 s ≥ elapsed since the last warning	AP_HAL::millis() - ekf_check_state.last_warn_time vs EKF_CHECK_WARNING_TIME (30 s)	Rate-limits the “EKF variance” GCS_SEVERITY_CRITICAL text to at most once per 30 s; stamps last_warn_time = millis().	<pre>// send message to gcs if ((AP_HAL::millis() - ekf_check_state.last_warn_time) > EKF_CHECK_WARNING_TIME) { gcs().send_text(MAV_SEVERITY_CRITICAL, "EKF variance"); ekf_check_state.last_warn_time = AP_HAL::millis(); }</pre>	INDIRECT/RELATIONAL (timing). [MULTI-CATEGORY w/ Table 5 #1 ekf_check]. [DUPLICATED] 30 s warn-throttle copied in ArduPlane/ekf_check.cpp.
7	ArduPlane/ekf_check.cpp	78-82	Plane::ekf_check() — GCS warnthrottle	EKF variance already flagged bad AND 30 s ≥ elapsed since the last warning	AP_HAL::millis() - ekf_check_state.last_warn_time vs EKF_CHECK_WARNING_TIME (30 s)	Rate-limits the “EKF variance” GCS text to at most once per 30 s; stamps last_warn_time = millis().	<pre>// send message to gcs if ((AP_HAL::millis() - ekf_check_state.last_warn_time) > EKF_CHECK_WARNING_TIME) { gcs().send_text(MAV_SEVERITY_CRITICAL, "EKF variance"); ekf_check_state.last_warn_time = AP_HAL::millis(); }</pre>	INDIRECT/RELATIONAL (timing). [DUPLICATED] mirrors ArduCopter/ekf_check.cpp:84-88.
8	ArduSub/failsafe.cpp	125-130	Sub::failsafe_ekf_check() — 2 s confirm hysteresis	EKF variance bad but not yet bad for a full 2 s	AP_HAL::millis() vs last_ekf_good_ms + 2000 (2 s clock-based confirmation window)	Suppresses the EKF failsafe (clears failsafe.ekf / ekf_bad and returns) until the EKF has been bad for 2 solid seconds, then trips → arming.disarm(EKFFAILSAFE).	<pre>// Bad EKF for 2 solid seconds triggers failsafe if (AP_HAL::millis() < last_ekf_good_ms + 2000) { failsafe.ekf = false; AP_Notify::flags.ekf_bad = false; return; }</pre>	INDIRECT/RELATIONAL (timing). [MULTI-CATEGORY w/ Table 5 #4]. Divergent hysteresis: Sub uses a clock-based 2 s window + 20 s GCS warn-throttle (L141), whereas Copter/Plane use a count-based EKF_CHECK_ITERATIONS_MAX (≈1 s) + 30 s warn-throttle.
9	ArduCopter/ekf_check.cpp	125-132	Copter::ekf_over_threshold() — dtfiltering	Each ekf_over_threshold() evaluation (nominally the 10 Hz ekf_check cadence)	AP_HAL::micros() delta dt = (now_us - → last_ekf_check_us) * 1e-6 (actual inter-call interval)	Applies the measured dt to the position/velocity variance low-pass filters (pos_variance_filt / vel_variance_filt); the effective filter cutoff tracks the real scheduler call interval rather than a fixed rate.	<pre>uint32_t now_us = AP_HAL::micros(); float dt = (now_us - last_ekf_check_us) * 1e-6f; // always update filtered values as this serves the vibration check as well position_var = pos_variance_filt.apply(position_var, dt); vel_var = vel_variance_filt.apply(vel_var, dt); last_ekf_check_us = now_us;</pre>	ArduPilot_PNT_Reference_Audit.pdf INDIRECT/RELATIONAL (timing). [MULTI-CATEGORY w/ Table 5 #1]. EKF check-rate <-> scheduler coupling (AAP 0.5.5): filter behaviour depends on scheduler timing.

#	File Path	Line(s)	Function/Class	Trigger Condition	PNT State Observed	Behavior Change Description	Code Snippet (5-10 lines)	Notes
10	ArduCopter/fence.cpp	8-16	Copter::fence_checks_async()	1 kHz async fence IO callback	AP_HAL::millis() (now) and AP_HAL::timeout_expired(fence_breaches.last_check_ms, now, 40U) (40 ms / 25 Hz gate)	Returns early until 40 ms have elapsed, throttling the geofence breach check to 25 Hz; the position-vs-fence observation itself is Table 4 #3.	<pre>void Copter::fence_checks_async() { const uint32_t now = AP_HAL::millis(); if (!AP_HAL::timeout_expired(fence_breaches.last_check_ms, now, 40U)) { // 25Hz update rate return; } if (fence_breaches.have_updates) {</pre>	INDIRECT/RELATIONAL (timing). [MULTI-CATEGORY w/ Table 4 #3] — timing side of the same function (F1). [DUPLICATED] vs Rover/fence.cpp (divergent rate).
11	Rover/fence.cpp	6-14	Rover::fence_checks_async()	1 kHz async fence IO callback	AP_HAL::millis() (now) and AP_HAL::timeout_expired(fence_breaches.last_check_ms, now, 100U) (100 ms / 10 Hz gate)	Returns early until 100 ms have elapsed, throttling the geofence breach check to 10 Hz.	<pre>void Rover::fence_checks_async() { const uint32_t now = AP_HAL::millis(); if (!AP_HAL::timeout_expired(fence_breaches.last_check_ms, now, 100U)) { // 10Hz update rate return; } if (fence_breaches.have_updates) {</pre>	INDIRECT/RELATIONAL (timing). [MULTI-CATEGORY w/ Table 4 #9]. [DUPLICATED] vs ArduCopter/fence.cpp:8-16 — divergent rate (40U/25 Hz vs 100U/10 Hz).

Table 6a — Indirect Timing dependencies (Layer 1 + Layer 2)

Layer 1 — Direct calls / called-by with typed dependency

Ref #	Component/Function	Directly Calls	Directly Called By	Dependency Type	Code Snippet
1	Copter::failsafe_check()	AP_HAL::micros(); AP_Scheduler::ticks(); motors->output_min(); motors->armed(false)	failsafe_check_static() <- hal.scheduler->register_timer_failsafe (1 kHz, system.cpp:87)	Function call	<pre>uint32_t tnow = AP_HAL::micros(); const uint16_t ticks = scheduler.ticks(); if (ticks != failsafe_last_ticks) { // the main loop is running, all is OK failsafe_last_ticks = ticks; failsafe_last_timestamp = tnow; if (in_failsafe) { in_failsafe = false;</pre>
2	Plane::failsafe_check()	micros(); AP_Scheduler::ticks(); rc().read_input(); SRV_Channels::set_output_scaled()	failsafe_check_static() <- hal.scheduler->register_timer_failsafe (1 kHz, system.cpp:103)	Function call	<pre>uint32_t tnow = micros(); const uint16_t ticks = scheduler.ticks(); if (ticks != last_ticks) { // the main loop is running, all is OK last_ticks = ticks; last_timestamp = tnow; in_failsafe = false; return;</pre>
3	Rover::failsafe_check()	AP_HAL::micros(); AP_Scheduler::ticks(); arming.disarm(CPUFAILSAFE)	failsafe_check_static() <- hal.scheduler->register_timer_failsafe (1 kHz, system.cpp:111)	Function call	<pre>const uint32_t tnow = AP_HAL::micros(); const uint16_t ticks = scheduler.ticks(); if (ticks != last_ticks) { // the main loop is running, all is OK last_ticks = ticks; last_timestamp = tnow; return; }</pre>
4	Sub::mainloop_failsafe_check()	AP_HAL::micros(); AP_Scheduler::ticks(); motors.output_min()	failsafe_check_static() <- hal.scheduler->register_timer_failsafe (1 kHz, system.cpp:71)	Function call	<pre>uint32_t tnow = AP_HAL::micros(); const uint16_t ticks = scheduler.ticks(); if (ticks != failsafe_last_ticks) { // the main loop is running, all is OK failsafe_last_ticks = ticks; failsafe_last_timestamp = tnow; if (in_failsafe) { in_failsafe = false;</pre>

Ref #	Component/Function	Directly Calls	Directly Called By	Dependency Type	Code Snippet
5	Blimp::failsafe_check()	AP_HAL::micros(); AP_Scheduler::ticks(); motors->output_min()	failsafe_check_static() <- hal.scheduler->register_timer_failsafe (1 kHz, system.cpp:59)	Function call	<pre>uint32_t tnow = AP_HAL::micros(); const uint16_t ticks = scheduler.ticks(); if (ticks != failsafe_last_ticks) { // the main loop is running, all is OK failsafe_last_ticks = ticks; failsafe_last_timestamp = tnow; if (in_failsafe) { in_failsafe = false; } }</pre>
6	Copter::ekf_check() (warn throttle)	AP_HAL::millis(); gcs().send_text()	Copter::ekf_check() bad-variance branch	Function call	<pre>// send message to gcs if ((AP_HAL::millis() - ekf_check_state.last_warn_time) > EKF_CHECK_WARNING_TIME) { gcs().send_text(MAV_SEVERITY_CRITICAL, "EKF variance"); ekf_check_state.last_warn_time = AP_HAL::millis(); }</pre>
7	Plane::ekf_check() (warn throttle)	AP_HAL::millis(); gcs().send_text()	Plane::ekf_check() bad-variance branch	Function call	<pre>// send message to gcs if ((AP_HAL::millis() - ekf_check_state.last_warn_time) > EKF_CHECK_WARNING_TIME) { gcs().send_text(MAV_SEVERITY_CRITICAL, "EKF variance"); ekf_check_state.last_warn_time = AP_HAL::millis(); }</pre>
8	Sub::failsafe_ekf_check() (2 s hysteresis)	AP_HAL::millis(); compares last_ekf_good_ms + 2000	Sub periodic scheduler	Function call	<pre>// Bad EKF for 2 solid seconds triggers failsafe if (AP_HAL::millis() < last_ekf_good_ms + 2000) { failsafe.ekf = false; AP_Notify::flags.ekf_bad = false; return; }</pre>
9	Copter::ekf_over_threshold() (dt)	AP_HAL::micros(); pos_variance_filt.apply(); vel_variance_filt.apply()	Copter::ekf_check()	Function call	<pre>uint32_t now_us = AP_HAL::micros(); float dt = (now_us - last_ekf_check_us) * 1e-6f; // always update filtered values as this serves the vibration check as well position_var = pos_variance_filt.apply(position_var, dt); vel_var = vel_variance_filt.apply(vel_var, dt); last_ekf_check_us = now_us;</pre>
10	Copter::fence_checks_async()	AP_HAL::millis(); AP_HAL::timeout_expired(...,40U)	1 kHz async IO process (hal.scheduler register_io_process)	Function call	<pre>void Copter::fence_checks_async() { const uint32_t now = AP_HAL::millis(); if (!AP_HAL::timeout_expired(fence_breaches.last_check_ms, now, 40U)) { // 25Hz update rate return; } if (fence_breaches.have_updates) {</pre>
11	Rover::fence_checks_async()	AP_HAL::millis(); AP_HAL::timeout_expired(...,100U)	1 kHz async IO process (hal.scheduler register_io_process)	Function call	<pre>void Rover::fence_checks_async() { const uint32_t now = AP_HAL::millis(); if (!AP_HAL::timeout_expired(fence_breaches.last_check_ms, now, 100U)) { // 10Hz update rate return; } if (fence_breaches.have_updates) {</pre>

Layer 2 — Full transitive chain to final consumer (hop count)

Ref #	PNT Origin	Full Dependency Chain	Final Consumer	Chain Depth	Notes
1	AP_HAL::micros() + AP_Scheduler tick counter	AP_Scheduler::ticks() / AP_HAL::micros() failsafe_check_static() [1 kHz timer-failsafe] → → Copter::failsafe_check() (>2 s stall) AP_Motors::output_min()/armed(false) →	AP_Motors (motor min / disarm)	3	[DUPLICATED] Main-loop-lockup watchdog replicated 5 ways — ArduCopter/failsafe.cpp, ArduPlane/failsafe.cpp, Rover/failsafe.cpp, ArduSub/failsafe.cpp (mainloop_failsafe_check) and Blimp/failsafe.cpp — each reads AP_Scheduler::ticks() + AP_HAL::micros(); divergent thresholds/actions (extraction risk).
2	AP_HAL::micros() + AP_Scheduler tick counter	AP_Scheduler::ticks() / micros() failsafe_check_static() Plane::failsafe_check() (>0.2 s stall) → → → SRV_Channels::set_output_scaled() servos_output() →	SRV_Channels (RC passthrough)	4	[DUPLICATED] watchdog; divergent 0.2 s trip + RC-passthrough action.
3	AP_HAL::micros() + AP_Scheduler tick counter	AP_Scheduler::ticks() / micros() failsafe_check_static() Rover::failsafe_check() (>0.2 s stall) → → → AP_Arming::disarm(CPUFAILSAFE)	AP_Arming (disarm)	3	[DUPLICATED] watchdog.
4	AP_HAL::micros() + AP_Scheduler tick counter	AP_Scheduler::ticks() / micros() failsafe_check_static() Sub::mainloop_failsafe_check() (>2 s → → stall) AP_Motors::output_min() →	AP_Motors (motor min)	3	[DUPLICATED] watchdog.
5	AP_HAL::micros() + AP_Scheduler tick counter	AP_Scheduler::ticks() / micros() failsafe_check_static() Blimp::failsafe_check() (>2 s stall) → → → AP_Motors::output_min()	AP_Motors (motor min)	3	[DUPLICATED] watchdog; Blimp observes scheduler timing state only (AAP 0.3.2).
6	AP_HAL::millis() wall clock	AP_HAL::millis() Copter::ekf_check() (>30 s since last_warn) gcs().send_text(“EKF variance”) → → GCS_MAVLINK →	GCS_MAVLINK (telemetry text)	3	[DUPLICATED] 30 s warn-throttle (Copter/Plane). [MULTI-CATEGORY w/ Table 5 #1].
7	AP_HAL::millis() wall clock	AP_HAL::millis() Plane::ekf_check() (>30 s since last_warn) gcs().send_text(“EKF variance”) → → → GCS_MAVLINK	GCS_MAVLINK (telemetry text)	3	[DUPLICATED] mirrors #6.
8	AP_HAL::millis() wall clock	AP_HAL::millis() Sub::failsafe_ekf_check() (EKF bad 2 s) failsafe.ekf = true → ≥ → → arming.disarm(EKFFAILSAFE)	AP_Arming (EKF failsafe disarm)	3	[MULTI-CATEGORY w/ Table 5 #4]. 2 s confirm + 20 s warn-throttle.
9	AP_HAL::micros() inter-call interval	AP_HAL::micros() dt pos/vel_variance_filt.apply(var, dt) ekf_over_threshold() → → → → Copter::ekf_check() failsafe decision	Copter::ekf_check() variance decision	4	EKF check-rate <-> scheduler coupling (AAP 0.5.5). [MULTI-CATEGORY w/ Table 5 #1].
10	AP_HAL::millis() wall clock	AP_HAL::millis() AP_HAL::timeout_expired(40U) gate Copter::fence_checks_async() → → → AC_Fence::check() fence_breaches latched main-loop fence failsafe → →	AC_Fence / fence failsafe	5	[MULTI-CATEGORY w/ Table 4 #3]. [DUPLICATED] vs Rover (40U/25 Hz).
11	AP_HAL::millis() wall clock	AP_HAL::millis() AP_HAL::timeout_expired(100U) gate Rover::fence_checks_async() → → → AC_Fence::check() fence_breaches latched →	AC_Fence / fence failsafe	4	ArduPilot_PNT_Reference_Audit.pdf [MULTI-CATEGORY w/ Table 4 #9]. [DUPLICATED] vs Copter (100U/10 Hz).

COVERAGE & EXPLICIT-ABSENCE REGISTER

Exhaustive survey ledger: every required surface is either catalogued above (with table/row references) or recorded here as a checked explicit absence. Non-inference discipline (no “handled elsewhere”). Indirect supporting-library register (AAP 0.3.1) — AC_Fence / AP_BattMonitor / AP_RCProtocol: non-PNT triggers whose failsafe action is PNT-gated, plus checked absences.

Surface/File	PNT Intersection	Detail/Predicate/Behavior	Evidence
Exhaustive survey ledger: every required surface is Indirect supporting-library register (AAP absences).	either catalogued above 0.3.1) — AC_Fence /	(with table/row references) or recorded here as a checked explicit absence. Non-inference AP_BattMonitor / AP_RCProtocol: non-PNT triggers whose failsafe	discipline (no “handled elsewhere”). action is PNT-gated, plus checked
libraries/AC_Fence/ (AC_Fence::check +	Catalogued	Reads vehicle position vs polygon / circle / altitude fence; position-dependent breach. Fully	libraries/AC_Fence/AC_Fence.cpp:679-843 · Table 4 #4-#8 /
check_fence_polygon / check_fence_circle)	✓	catalogued in the numbered tables (not an absence).	Table 4a
libraries/AP_BattMonitor/ (58 source files; AP_BattMonitor::check_failsafes)	[ABSENT] checked	Directory-wide token sweep found ZERO PNT reads (no ahrs. / gps. / get_location / position_ok / get_velocity_NED). Failsafe trigger is voltage / capacity only; PNT enters at the vehicle action layer, not here.	libraries/AP_BattMonitor/AP_BattMonitor.cpp:781 (check_failsafes) · action cross-ref ArduCopter/events.cpp:99-121
libraries/AP_RCProtocol/ (RC failsafe flag) ArduCopter/events.cpp (handle_battery_failsafe / radio /	[ABSENT] checked PNT-reactive	Token sweep found ZERO PNT reads. Failsafe is a boolean signal-present flag (failsafe_active); no position / nav / PNT-timing observation. PNT enters at the vehicle action layer. Non-PNT trigger (battery / RC / GCS) but the chosen failsafe action (RTL / SmartRTL) is gated	libraries/AP_RCProtocol/AP_RCProtocol.h:126-130 (failsafe_active) · action cross-ref ArduCopter/events.cpp:99-121 ArduCopter/events.cpp:99-121 (do_failsafe_action
GCS FS action select) Cross-vehicle indirect safety-surface register	✓ (F16) — Rover /	on position_ok and degrades to LAND when position is invalid — the relational PNT site for batt / RC / GCS failsafe. ArduSub failsafe, fence, and mode surfaces, each resolved to its	BATTERY_FAILSAFE) · gate Table 4 #2 catalogued numbered-table row.
Rover/failsafe.cpp (Rover::failsafe_check main-loop	Catalogued	Main-loop-lockup watchdog reading scheduler ticks + micros; 0.2 s trip	Rover/failsafe.cpp:18-31 · Table 6 #3 / Table 6a #3
watchdog)	✓	→ arming.disarm(CPUFAILSAFE). Catalogued, not an absence.	
Rover/fence.cpp (Rover::fence_checks_async)	Catalogued	Geofence position-vs-boundary check on a 100 ms (10 Hz) timing gate; position-dependent	Rover/fence.cpp:6-14 · Table 6 #11 / Table 6a #11

Surface/File	PNT Intersection	Detail/Predicate/Behavior	Evidence
	✓	breach. Catalogued.	
ArduSub/failsafe.cpp (Sub::failsafe_ekf_check)	Catalogued	EKF-variance health watchdog with 2 s confirm + 20 s warn hysteresis. Catalogued.	ArduSub/failsafe.cpp:100-150 · Table 5 #4 / Table 6 #8
	✓		
ArduSub/failsafe.cpp (Sub::mainloop_failsafe_check)	Catalogued	Main-loop-lockup watchdog (scheduler ticks + micros; 2 s motors.output_min). Catalogued.	ArduSub/failsafe.cpp:29-55 · Table 6 #4 / Table 6a #4
	✓	→	
ArduSub/mode*.cpp (12 files)	Catalogued	All 12 Sub mode files enumerated with per-file requires_GPS gating and checked absences in	ArduSub/mode.cpp:89-90 (set_mode gate) · Sub
AntennaTracker & Blimp PNT-intersection checked absences. AntennaTracker/tracking.cpp	✓ register (F18; AAP PNT consumer	the Sub mode-gating matrix below. 0.3.2 — included only where they observe / consume PNT): presence Reads vehicle.location and vehicle.vel and dead-reckons location_estimate (offset by vel times	mode-gating matrix (this appendix) sites, per-mode gating, and AntennaTracker/tracking.cpp:7-26
(Tracker::update_vehicle_pos_estimate) AntennaTracker/mode_auto.cpp + mode_guided.cpp	✓ PNT consumer	dt) to point the antenna. Direct PNT consumption (presence, not absence). Auto / Guided pointing modes consume the tracked vehicle position / bearing to drive pitch and	AntennaTracker/mode_auto.cpp:5-10 · consumes
	✓	yaw. PNT-reactive.	tracking.cpp:7-26
AntennaTracker/mode_manual.cpp + mode_scan.cpp + mode_servotest.cpp Blimp/mode.cpp (Blimp::set_mode position gate)	[ABSENT] checked Gate enforcer	Manual / Scan / Servotest drive the antenna by stick / pattern / test only; token sweep found ZERO vehicle-position reads. Checked absence. Rejects PNT modes when position invalid: requires_GPS() and not blimp.position_ok(). Same	AntennaTracker/mode_servotest.cpp (0 PNT tokens) · mode_scan.cpp / mode_manual.cpp (0) Blimp/mode.cpp:78-79
Blimp/mode_velocity.cpp (ModeVelocity)	✓ Position-gated	gate idiom as Copter / Sub. requires_GPS()==true needs a valid position / velocity estimate to enter.	Blimp/mode.h:181 (requires_GPS=true) · gate
Blimp/mode_loiter.cpp (ModeLoiter)	✓ Position-gated	→ requires_GPS()==true position-gated.	Blimp/mode.cpp:78-79 Blimp/mode.h:226 (requires_GPS=true) · gate
Blimp/mode_rtl.cpp (ModeRTL)	✓ Position-gated	→ requires_GPS()==true position-gated.	Blimp/mode.cpp:78-79 ArduPilot_PNT_Reference_Audit.pdf Blimp/mode.h:315 (requires_GPS=true) · gate
	✓	→	Blimp/mode.cpp:78-79
Blimp/mode_manual.cpp (ModeManual)	[ABSENT] checked	requires_GPS()==false no position gate; manual stick control. Checked absence. →	Blimp/mode.h:138 (requires_GPS=false)
Blimp/mode_land.cpp (ModeLand)	[ABSENT] checked	requires_GPS()==false no horizontal-position gate (descent uses non-PNT references). → Checked absence for horizontal PNT.	Blimp/mode.h:271 (requires_GPS=false)
Blimp/failsafe.cpp (Blimp::failsafe_check watchdog)	Catalogued	Main-loop-lockup watchdog (scheduler ticks + micros; 2 s output_min). Catalogued.	Blimp/failsafe.cpp:37-45 · Table 6 #5 / Table 6a #5
External-boundary dependency-edge register PNT edge or publisher site, mirroring the none silently omitted.	✓ (AAP 0.4.1 — .gitmodules mavlink / DroneCAN	→ PNT-relevant vendored submodule boundaries): each edge treatment. Non-inference discipline — every .gitmodules	boundary resolved to its catalogued PNT edge is accounted for explicitly,
modules/mavlink (MAVLink message definitions)	Catalogued (edge)	GPS_INPUT ingress (AP_GPS_MAV Source backend) and NAV_CONTROLLER_OUTPUT	.gitmodules:7-9 · Table 1 (AP_GPS_MAV.cpp:67-75
	✓	egress (navigation-health reporting to the GCS). PNT telemetry boundary — catalogued, not an absence.	handle_msg() · Table 5 (ArduPlane/GCS_MAVLink_Plane.cpp:204-212 send_nav_controller_output)
modules/DroneCAN/DSDL + libcanard (CAN-bus GNSS	Catalogued (edge)	DroneCAN / UAVCAN Fix2 GNSS messages decoded into GPS_State. Positioning-source	.gitmodules:19-32 · Table 1
message definitions)	✓	boundary — catalogued, not an absence.	(AP_GPS_DroneCAN.cpp:388-396 handle_fix2_msg)
modules/ChibiOS (RTOS low-level clock / tick primitives)	Catalogued (edge)	Board RTOS timer backs the AP_HAL::micros() / millis() monotonic-time primitives. Timing	.gitmodules:13-15 · Table 3 #1 (AP_HAL/system.h:15-20) ·
modules/Micro-XRCE-DDS-Client + modules/Micro-CDR	✓ PNT publisher (edge)	boundary — named explicitly here as the board-timer origin of Table 3a #1. AP_DDS (AP_DDS_ENABLED=1 by default) reads PNT state and publishes it over the	Table 3a #1 (RTOS / board-hardware-timer origin) .gitmodules:34-40 ·
(DDS transport, AP_DDS ROS2) → ArduPlane vehicle-dir indirect glue register to / consumes PNT state, mirroring the catalogued ArduPlane/events.cpp (Plane::handle_battery_failsafe /	✓ (vehicle-symmetry ArduCopter PNT-reactive	Micro-XRCE-DDS / Micro-CDR transport to ROS2: sensor_msgs/NavSatFix from gps.location() and geographic_msgs/GeoPoseStamped from ahrs.get_location(). PNT-egress boundary — same edge treatment as mavlink / DroneCAN. completeness, AAP 0.3.1 / 0.6.1): ArduPlane failsafe-action and twins. Non-PNT trigger (battery / radio / GCS) but the chosen failsafe action is PNT-gated:	libraries/AP_DDS/AP_DDS_Client.cpp:268-270 (NavSatFix gps.location() · :577-584 (GeoPose ahrs.get_location() · ← ← AP_DDS_config.h:11 (#define AP_DDS_ENABLED 1) altitude-management glue that reacts ArduPlane/events.cpp:290-300 (have_position / current_loc
failsafe_long_on_event / failsafe_short_on_event) ArduPlane/altitude.cpp (Plane::setup_alt_slope /	✓ PNT consumer (Sink)	handle_battery_failsafe reads plane.have_position and plane.current_loc to choose land-sequence continuation vs RTL. Direct Plane analog of the catalogued ArduCopter/events.cpp. Positioning Sink: re-reads already-catalogued PNT state — current_loc.get_distance() /	land-sequence gate; def @265) · symmetric with ArduCopter/events.cpp:99-121 · gate ArduPlane/mode.cpp:144 ArduPlane/altitude.cpp:48-55 (setup_alt_slope current_loc
check_home_alt_change) AP_AHRS backend register (Navigation-hub navigation accessors behind the hub.	✓ completeness): the	line_path_proportion() (AHRS get_location, Table 2) and ahrs.get_home() — to derive the glide slope and altitude target. Consumes, does not produce, PNT. AP_AHRS frontend hub is catalogued in Table 2 / 2a; each concrete	distance) · :31-43 (check_home_alt_change ahrs.get_home() · :109 & :202 (current_loc.alt) backend overrides the

Surface/File	PNT Intersection	Detail/Predicate/Behavior	Evidence
AP_AHRS_DCM (libraries/AP_AHRS/AP_AHRS_DCM.cpp)	Catalogued	Legacy DCM estimator backend; overrides get_location / groundspeed. Cited in the numbered	libraries/AP_AHRS/AP_AHRS_DCM.cpp · Table 2 / Table 2a
AP_AHRS_External	✓ PNT backend	Navigation tables alongside the hub. External-AHRS backend feeding the hub from an external INS; overrides the position accessor	(AHRS hub + DCM) libraries/AP_AHRS/AP_AHRS_External.cpp:128 (get_origin)
(libraries/AP_AHRS/AP_AHRS_External.cpp) AP_AHRS_SIM (libraries/AP_AHRS/AP_AHRS_SIM.cpp)	✓ PNT backend	(get_origin). Navigation Source backend behind the AP_AHRS frontend. SITL simulation backend supplying simulated navigation state to the hub; overrides get_location	· AP_AHRS_External.h:30 (class : public AP_AHRS_Backend) libraries/AP_AHRS/AP_AHRS_SIM.cpp:7 (get_location) ·
	✓	/ get_origin. Navigation Source backend behind the AP_AHRS frontend.	:194 (get_origin) · AP_AHRS_SIM.h:37 (class : public AP_AHRS_Backend) ArduPilot_PNT_Reference_Audit.pdf
Copter flight modes PNT-gating matrix — 29 files (21 PNT-relevant, 8 checked-absence). Predicate: requires_GPS().			
mode.cpp (Copter::set_mode() position gate)	Gate enforcer	new_flightmode->requires_GPS() && !copter.position_ok() rejects PNT modes when position	ArduCopter/mode.cpp:303-307 · Table 4 #2
	✓	invalid	
mode_acro.cpp (ModeAcro)	[ABSENT] checked	requires_GPS()==false and zero position/nav/velocity tokens — manual/attitude mode; no PNT dependence (checked absence).	ArduCopter/mode.h:428 · gate Table 4 #2 (ArduCopter/mode.cpp:303-307)
mode_acro_heli.cpp (ModeAcro_Heli) mode_althold.cpp (ModeAltHold)	[ABSENT] checked PNT (no horiz-GPS gate)	requires_GPS()==false and zero position/nav/velocity tokens — manual/attitude mode; no PNT dependence (checked absence). requires_GPS()==false, but consumes PNT here (pos_control x9; e.g. altitude pos_control /	ArduCopter/mode.h:428 · gate Table 4 #2 (ArduCopter/mode.cpp:303-307) ArduCopter/mode.h:484 · use
mode_auto.cpp (ModeAuto)	✓ Position-gated	optical-flow / optional position). requires_GPS()==true (conditional: all sub-modes except NAV_ATTITUDE_TIME) — PNT	ArduCopter/mode_althold.cpp:13 · gate Table 4 #2 (ArduCopter/mode.cpp:303-307) ArduCopter/mode_auto.cpp:178 · use
	✓	position/nav required to enter; behaviour gated.	ArduCopter/mode_auto.cpp:42 · gate Table 4 #2 (ArduCopter/mode.cpp:303-307)
mode_autorotate.cpp (ModeAutorotate) mode_autotune.cpp (ModeAutoTune)	[ABSENT] checked PNT (no horiz-GPS gate)	requires_GPS()==false and zero position/nav/velocity tokens — manual/attitude mode; no PNT dependence (checked absence). requires_GPS()==false, but consumes PNT here (pos_control x5; e.g. altitude pos_control /	ArduCopter/mode.h:2047 · gate Table 4 #2 (ArduCopter/mode.cpp:303-307) ArduCopter/mode.h:829 · use
mode_avoid_adsb.cpp (ModeAvoidADSB)	✓ Position-gated	optical-flow / optional position). requires_GPS()==true — PNT position/nav required to enter; behaviour gated.	ArduCopter/mode_autotune.cpp:31 · gate Table 4 #2 (ArduCopter/mode.cpp:303-307) ArduCopter/mode.h:1888 · gate Table 4 #2
mode_brake.cpp (ModeBrake)	✓ Position-gated	requires_GPS()==true — PNT position/nav required to enter; behaviour gated.	(ArduCopter/mode.cpp:303-307) ArduCopter/mode.h:854 · use
mode_circle.cpp (ModeCircle)	✓ Position-gated	requires_GPS()==true — PNT position/nav required to enter; behaviour gated.	ArduCopter/mode_brake.cpp:13 · gate Table 4 #2 (ArduCopter/mode.cpp:303-307) ArduCopter/mode.h:884 · use
mode_drift.cpp (ModeDrift)	✓ Position-gated	requires_GPS()==true — PNT position/nav required to enter; behaviour gated.	ArduCopter/mode_circle.cpp:16 · gate Table 4 #2 (ArduCopter/mode.cpp:303-307) ArduCopter/mode.h:914 · use ArduCopter/mode_drift.cpp:54
	✓		· gate Table 4 #2 (ArduCopter/mode.cpp:303-307)
mode_flip.cpp (ModeFlip) mode_flowhold.cpp (ModeFlowHold)	[ABSENT] checked PNT (no horiz-GPS gate)	requires_GPS()==false and zero position/nav/velocity tokens — manual/attitude mode; no PNT dependence (checked absence). requires_GPS()==false, but consumes PNT here (pos_control x12; e.g. altitude pos_control /	ArduCopter/mode.h:942 · gate Table 4 #2 (ArduCopter/mode.cpp:303-307) ArduCopter/mode.h:989 · use
mode_follow.cpp (ModeFollow)	✓ Position-gated	optical-flow / optional position). requires_GPS()==true — PNT position/nav required to enter; behaviour gated.	ArduCopter/mode_flowhold.cpp:92 · gate Table 4 #2 (ArduCopter/mode.cpp:303-307) ArduCopter/mode.h:1918 · use
mode_guided.cpp (ModeGuided)	✓ Position-gated	requires_GPS()==true — PNT position/nav required to enter; behaviour gated.	ArduCopter/mode_follow.cpp:33 · gate Table 4 #2 (ArduCopter/mode.cpp:303-307) ArduCopter/mode.h:1077 · use
mode_guided_custom.cpp (ModeGuidedCustom)	✓ Position-gated	requires_GPS()==true — PNT position/nav required to enter; behaviour gated.	ArduCopter/mode_guided.cpp:77 · gate Table 4 #2 (ArduCopter/mode.cpp:303-307) ArduCopter/mode.h:1077 · gate Table 4 #2
	✓		(ArduCopter/mode.cpp:303-307)
mode_guided_nogps.cpp (ModeGuidedNoGPS) mode_land.cpp (ModeLand)	[ABSENT] checked PNT (no horiz-GPS gate)	requires_GPS()==false and zero position/nav/velocity tokens — manual/attitude mode; no PNT dependence (checked absence). requires_GPS()==false, but consumes PNT here (position_ok x9; e.g. altitude pos_control /	ArduCopter/mode.h:1253 · gate Table 4 #2 (ArduCopter/mode.cpp:303-307) ArduCopter/mode.h:1277 · use ArduCopter/mode_land.cpp:7
mode_loiter.cpp (ModeLoiter)	✓ Position-gated	optical-flow / optional position). requires_GPS()==true — PNT position/nav required to enter; behaviour gated.	· gate Table 4 #2 (ArduCopter/mode.cpp:303-307) ArduPilot_PNT_Reference_Audit.pdf ArduCopter/mode.h:1323 · use

Surface/File	PNT Intersection	Detail/Predicate/Behavior	Evidence
mode_poshold.cpp (ModePosHold)	✓ Position-gated	requires_GPS()==true — PNT position/nav required to enter; behaviour gated.	ArduCopter/mode_loiter.cpp:17 · gate Table 4 #2 (ArduCopter/mode.cpp:303-307) ArduCopter/mode.h:1373 · use
mode_rtl.cpp (ModeRTL)	✓ Position-gated	requires_GPS()==true — PNT position/nav required to enter; behaviour gated.	ArduCopter/mode_poshold.cpp:28 · gate Table 4 #2 (ArduCopter/mode.cpp:303-307) ArduCopter/mode.h:1461 · use ArduCopter/mode_rtl.cpp:21
mode_smart_rtl.cpp (ModeSmartRTL)	✓ Position-gated	requires_GPS()==true — PNT position/nav required to enter; behaviour gated.	· gate Table 4 #2 (ArduCopter/mode.cpp:303-307) ArduCopter/mode.h:1577 · use
mode_sport.cpp (ModeSport)	✓ PNT (no horiz-GPS gate)	requires_GPS()==false, but consumes PNT here (pos_control ×9; e.g. altitude pos_control /	ArduCopter/mode_smart_rtl.cpp:16 · gate Table 4 #2 (ArduCopter/mode.cpp:303-307) ArduCopter/mode.h:1637 · use
	✓	optical-flow / optional position).	ArduCopter/mode_sport.cpp:13 · gate Table 4 #2 (ArduCopter/mode.cpp:303-307)
mode_stabilize.cpp (ModeStabilize)	[ABSENT] checked	requires_GPS()==false and zero position/nav/velocity tokens — manual/attitude mode; no PNT dependence (checked absence).	ArduCopter/mode.h:1664 · gate Table 4 #2 (ArduCopter/mode.cpp:303-307)
mode_stabilize_heli.cpp (ModeStabilize_Heli) mode_systemid.cpp (ModeSystemId)	[ABSENT] checked PNT (no horiz-GPS gate)	requires_GPS()==false and zero position/nav/velocity tokens — manual/attitude mode; no PNT dependence (checked absence). requires_GPS()==false, but consumes PNT here (pos_control ×22; e.g. altitude pos_control /	ArduCopter/mode.h:1664 · gate Table 4 #2 (ArduCopter/mode.cpp:303-307) ArduCopter/mode.h:1712 · use
mode_throw.cpp (ModeThrow)	✓ Position-gated	optical-flow / optional position). requires_GPS()==true — PNT position/nav required to enter; behaviour gated.	ArduCopter/mode_systemid.cpp:115 · gate Table 4 #2 (ArduCopter/mode.cpp:303-307) ArduCopter/mode.h:1791 · use
	✓		ArduCopter/mode_throw.cpp:23 · gate Table 4 #2 (ArduCopter/mode.cpp:303-307)
mode_turtle.cpp (ModeTurtle) mode_zigzag.cpp (ModeZigZag)	[ABSENT] checked Position-gated	requires_GPS()==false and zero position/nav/velocity tokens — manual/attitude mode; no PNT dependence (checked absence). requires_GPS()==true — PNT position/nav required to enter; behaviour gated.	ArduCopter/mode.h:1849 · gate Table 4 #2 (ArduCopter/mode.cpp:303-307) ArduCopter/mode.h:1967 · use
	✓		ArduCopter/mode_zigzag.cpp:75 · gate Table 4 #2 (ArduCopter/mode.cpp:303-307)
Plane flight modes PNT-gating matrix — 26 files (20 PNT-relevant, 6 checked-absence). Predicate: does_auto_navigation().			
mode.cpp (Plane::set_mode() / mode enter)	Gate enforcer	does_auto_navigation() consulted; nav modes consume EKF/AHRS nav state	ArduPlane/mode.cpp:144
mode_LoiterAltQLand.cpp (ModeLoiterAltQLand)	✓ Position-gated	does_auto_navigation()==true — PNT position/nav required to enter; behaviour gated.	ArduPlane/mode.h:427 · use
	✓		ArduPlane/mode_LoiterAltQLand.cpp:8 · gate ArduPlane/mode.cpp:144
mode_acro.cpp (ModeAcro) mode_auto.cpp (ModeAuto)	[ABSENT] checked Position-gated	does_auto_navigation()==false and zero position/nav/velocity tokens — manual/attitude mode; no PNT dependence (checked absence). does_auto_navigation()==true — PNT position/nav required to enter; behaviour gated.	gate ArduPlane/mode.cpp:144 ArduPlane/mode_auto.cpp:137 · use
mode_autoland.cpp (ModeAutoLand)	✓ Position-gated	does_auto_navigation()==true — PNT position/nav required to enter; behaviour gated.	ArduPlane/mode_auto.cpp:10 · gate ArduPlane/mode.cpp:144 ArduPlane/mode.h:957 · use
	✓		ArduPlane/mode_autoland.cpp:79 · gate ArduPlane/mode.cpp:144
mode_autotune.cpp (ModeAutoTune) mode_avoidADSB.cpp (ModeAvoidADSB)	[ABSENT] checked PNT (no horiz-GPS gate)	does_auto_navigation()==false and zero position/nav/velocity tokens — manual/attitude mode; no PNT dependence (checked absence). does_auto_navigation()==false, but consumes PNT here (update_loiter ×1; e.g. altitude	gate ArduPlane/mode.cpp:144 ArduPilot_PNT_Reference_Audit.pdf use ArduPlane/mode_avoidADSB.cpp:19 · gate
mode_circle.cpp (ModeCircle)	✓ Position-gated	pos_control / optical-flow / optional position). does_auto_navigation()==true — PNT position/nav required to enter; behaviour gated.	ArduPlane/mode.cpp:144 ArduPlane/mode.h:399 · use ArduPlane/mode_circle.cpp:7 ·
mode_cruise.cpp (ModeCruise)	✓ PNT (no horiz-GPS gate)	does_auto_navigation()==false, but consumes PNT here (prev_WP_loc ×4; e.g. altitude	gate ArduPlane/mode.cpp:144 use ArduPlane/mode_cruise.cpp:83 · gate
	✓	pos_control / optical-flow / optional position).	ArduPlane/mode.cpp:144
mode_fbwa.cpp (ModeFBWA)	[ABSENT] checked	does_auto_navigation()==false and zero position/nav/velocity tokens — manual/attitude mode; no PNT dependence (checked absence).	gate ArduPlane/mode.cpp:144
mode_fbwb.cpp (ModeFBWB) mode_guided.cpp (ModeGuided)	[ABSENT] checked Position-gated	does_auto_navigation()==false and zero position/nav/velocity tokens — manual/attitude mode; no PNT dependence (checked absence). does_auto_navigation()==true — PNT position/nav required to enter; behaviour gated.	gate ArduPlane/mode.cpp:144 ArduPlane/mode.h:360 · use

Surface/File	PNT Intersection	Detail/Predicate/Behavior	Evidence
mode_loiter.cpp (ModeLoiter)	✓ Position-gated	does_auto_navigation()=true — PNT position/nav required to enter; behaviour gated.	ArduPlane/mode_guided.cpp:11 · gate ArduPlane/mode.cpp:144 ArduPlane/mode.h:427 · use ArduPlane/mode_loiter.cpp:7 ·
mode_manual.cpp (ModeManual)	✓ PNT (no horiz-GPS gate)	does_auto_navigation()=false, but consumes PNT here (quadplane. ×1; e.g. altitude	gate ArduPlane/mode.cpp:144 use ArduPlane/mode_manual.cpp:26 · gate
mode_qacro.cpp (ModeQAcro)	✓ PNT (no horiz-GPS gate)	pos_control / optical-flow / optional position). does_auto_navigation()=false, but consumes PNT here (quadplane. ×16; e.g. altitude	ArduPlane/mode.cpp:144 use ArduPlane/mode_qacro.cpp:8 · gate
mode_qautotune.cpp (ModeQAutotune)	✓ PNT (no horiz-GPS gate)	pos_control / optical-flow / optional position). does_auto_navigation()=false, but consumes PNT here (quadplane. ×4; e.g. altitude	ArduPlane/mode.cpp:144 use ArduPlane/mode_qautotune.cpp:11 · gate
mode_qhover.cpp (ModeQHover)	✓ PNT (no horiz-GPS gate)	pos_control / optical-flow / optional position). does_auto_navigation()=false, but consumes PNT here (pos_control ×13; e.g. altitude	ArduPlane/mode.cpp:144 use ArduPlane/mode_qhover.cpp:9 · gate
mode_qland.cpp (ModeQLand)	✓ PNT (no horiz-GPS gate)	pos_control / optical-flow / optional position). does_auto_navigation()=false, but consumes PNT here (quadplane. ×5; e.g. altitude	ArduPlane/mode.cpp:144 use ArduPlane/mode_qland.cpp:9 · gate
mode_qloiter.cpp (ModeQLoiter)	✓ PNT (no horiz-GPS gate)	pos_control / optical-flow / optional position). does_auto_navigation()=false, but consumes PNT here (loiter_nav ×52; e.g. altitude	ArduPlane/mode.cpp:144 use ArduPlane/mode_qloiter.cpp:9 · gate
mode_qrtl.cpp (ModeQRTL)	✓ PNT (no horiz-GPS gate)	pos_control / optical-flow / optional position). does_auto_navigation()=false, but consumes PNT here (quadplane. ×47; e.g. altitude	ArduPlane/mode.cpp:144 use ArduPlane/mode_qrtl.cpp:10 · gate
mode_qstabilize.cpp (ModeQStabilize)	✓ PNT (no horiz-GPS gate)	pos_control / optical-flow / optional position). does_auto_navigation()=false, but consumes PNT here (quadplane. ×19; e.g. altitude	ArduPlane/mode.cpp:144 use ArduPlane/mode_qstabilize.cpp:8 · gate
mode_rtl.cpp (ModeRTL)	✓ Position-gated	pos_control / optical-flow / optional position). does_auto_navigation()=true — PNT position/nav required to enter; behaviour gated.	ArduPlane/mode.cpp:144 ArduPlane/mode.h:507 · use ArduPlane/mode_rtl.cpp:6 ·
	✓		gate ArduPlane/mode.cpp:144
mode_stabilize.cpp (ModeStabilize) mode_takeoff.cpp (ModeTakeoff)	[ABSENT] checked Position-gated	does_auto_navigation()=false and zero position/nav/velocity tokens — manual/attitude mode; no PNT dependence (checked absence). does_auto_navigation()=true — PNT position/nav required to enter; behaviour gated.	gate ArduPlane/mode.cpp:144 ArduPlane/mode.h:909 · use
mode_thermal.cpp (ModeThermal)	✓ Position-gated	does_auto_navigation()=true — PNT position/nav required to enter; behaviour gated.	ArduPlane/mode_takeoff.cpp:74 · gate ArduPlane/mode.cpp:144 ArduPlane/mode.h:1025 · use
	✓		ArduPlane/mode_thermal.cpp:16 · gate ArduPlane/mode.cpp:144
mode_training.cpp (ModeTraining)	[ABSENT] checked	does_auto_navigation()=false and zero position/nav/velocity tokens — manual/attitude mode; no PNT dependence (checked absence).	gate ArduPlane/mode.cpp:144
Rover drive modes PNT-gating matrix — 14 files (11 PNT-relevant, 3 checked-absence). Predicate: requires_position().			
mode.cpp (Mode::enter() position gate)	Gate enforcer	rover.ekf_position_ok() && requires_position()/requires_velocity() gate entry	ArduPilot_PNT_Reference_Audit.pdf Rover/mode.cpp:32-38
mode_acro.cpp (ModeAcro)	✓ PNT (no horiz-GPS gate)	requires_position()=false, but consumes PNT here (wp_nav ×1; e.g. altitude pos_control /	Rover/mode.h:235 · use Rover/mode_acro.cpp:37 · gate
mode_auto.cpp (ModeAuto)	✓ Position-gated	optical-flow / optional position). requires_position()=true — PNT position/nav required to enter; behaviour gated.	Rover/mode.cpp:32-38 use Rover/mode_auto.cpp:14 · gate Rover/mode.cpp:32-38
mode_circle.cpp (ModeCircle)	✓ Position-gated	requires_position()=true — PNT position/nav required to enter; behaviour gated.	use Rover/mode_circle.cpp:85 · gate Rover/mode.cpp:32-38
mode_dock.cpp (ModeDock)	✓ Position-gated	requires_position()=true — PNT position/nav required to enter; behaviour gated.	use Rover/mode_dock.cpp:75 · gate Rover/mode.cpp:32-38
mode_follow.cpp (ModeFollow)	✓ Position-gated	requires_position()=true — PNT position/nav required to enter; behaviour gated.	use Rover/mode_follow.cpp:12 · gate Rover/mode.cpp:32-38
mode_guided.cpp (ModeGuided)	✓ Position-gated	requires_position()=true — PNT position/nav required to enter; behaviour gated.	use Rover/mode_guided.cpp:15 · gate
	✓		Rover/mode.cpp:32-38
mode_hold.cpp (ModeHold) mode_loiter.cpp (ModeLoiter)	[ABSENT] checked Position-gated	requires_position()=false and zero position/nav/velocity tokens — manual/attitude mode; no PNT dependence (checked absence). requires_position()=true — PNT position/nav required to enter; behaviour gated.	Rover/mode.h:630 · gate Rover/mode.cpp:32-38 use Rover/mode_loiter.cpp:6 · gate Rover/mode.cpp:32-38
	✓		
mode_manual.cpp (ModeManual) mode_rtl.cpp (ModeRTL)	[ABSENT] checked Position-gated	requires_position()=false and zero position/nav/velocity tokens — manual/attitude mode; no PNT dependence (checked absence). requires_position()=true — PNT position/nav required to enter; behaviour gated.	Rover/mode.h:681 · gate Rover/mode.cpp:32-38 use Rover/mode_rtl.cpp:11 · gate Rover/mode.cpp:32-38
mode_simple.cpp (ModeSimple)	✓ Position-gated	requires_position()=true — PNT position/nav required to enter; behaviour gated.	gate Rover/mode.cpp:32-38

Surface/File	PNT Intersection	Detail/Predicate/Behavior	Evidence
mode_smart_rtl.cpp (ModeSmartRTL)	✓ Position-gated	requires_position()=true — PNT position/nav required to enter; behaviour gated.	use Rover/mode_smart_rtl.cpp:17 · gate
	✓		Rover/mode.cpp:32-38
mode_steering.cpp (ModeSteering)	[ABSENT] checked	requires_position()=false and zero position/nav/velocity tokens — manual/attitude mode; no PNT dependence (checked absence).	Rover/mode.h:785 · gate Rover/mode.cpp:32-38
Sub flight modes PNT-gating matrix — 12 files (7 PNT-relevant, 5 checked-absence). Predicate: requires_GPS().			
mode.cpp (Sub::set_mode() position gate)	Gate enforcer	new_flightmode->requires_GPS() && !sub.position_ok() rejects PNT modes	ArduSub/mode.cpp:89-90
	✓		
mode_acro.cpp (ModeAcro)	[ABSENT] checked	requires_GPS()=false and zero position/nav/velocity tokens — manual/attitude mode; no PNT dependence (checked absence).	ArduSub/mode.h:155 · gate ArduSub/mode.cpp:89-90
mode_althold.cpp (ModeAlthold) mode_auto.cpp (ModeAuto)	[ABSENT] checked Position-gated	requires_GPS()=false and zero position/nav/velocity tokens — manual/attitude mode; no PNT dependence (checked absence). requires_GPS()=true — PNT position/nav required to enter; behaviour gated.	ArduSub/mode.h:201 · gate ArduSub/mode.cpp:89-90 ArduSub/mode.h:314 · use ArduSub/mode_auto.cpp:11 ·
mode_circle.cpp (ModeCircle)	✓ Position-gated	requires_GPS()=true — PNT position/nav required to enter; behaviour gated.	gate ArduSub/mode.cpp:89-90 ArduSub/mode.h:382 · use ArduSub/mode_circle.cpp:10 ·
mode_guided.cpp (ModeGuided)	✓ Position-gated	requires_GPS()=true — PNT position/nav required to enter; behaviour gated.	gate ArduSub/mode.cpp:89-90 ArduSub/mode.h:260 · use ArduSub/mode_guided.cpp:34 ·
	✓		gate ArduSub/mode.cpp:89-90
mode_manual.cpp (ModeManual)	[ABSENT] checked	requires_GPS()=false and zero position/nav/velocity tokens — manual/attitude mode; no PNT dependence (checked absence).	ArduSub/mode.h:132 · gate ArduSub/mode.cpp:89-90
mode_motordetect.cpp (ModeMotordetect) mode_poshold.cpp (ModePoshold)	[ABSENT] checked Position-gated	requires_GPS()=false and zero position/nav/velocity tokens — manual/attitude mode; no PNT dependence (checked absence). requires_GPS()=true — PNT position/nav required to enter; behaviour gated.	ArduSub/mode.h:427 · gate ArduSub/mode.cpp:89-90 ArduSub/mode.h:355 · use ArduSub/mode_poshold.cpp:13 ·
	✓		gate ArduSub/mode.cpp:89-90
mode_stabilize.cpp (ModeStabilize) mode_surface.cpp (ModeSurface)	[ABSENT] checked PNT (no horiz-GPS gate)	requires_GPS()=false and zero position/nav/velocity tokens — manual/attitude mode; no PNT dependence (checked absence). requires_GPS()=false, but consumes PNT here (wp_nav ×1; e.g. altitude pos_control /	ArduSub/mode.h:178 · gate ArduSub/mode.cpp:89-90 ArduSub/mode.h:404 · use ArduSub/mode_surface.cpp:54 ·
mode_surftrak.cpp (ModeSurftrak)	✓ PNT (no horiz-GPS gate)	optical-flow / optional position). requires_GPS()=false, but consumes PNT here (pos_control ×3; e.g. altitude pos_control /	gate ArduSub/mode.cpp:89-90 ArduPilot_PNT_Reference_Audit.pdf ArduSub/mode.h:201 · use ArduSub/mode_surftrak.cpp:91 ·
	✓	optical-flow / optional position).	gate ArduSub/mode.cpp:89-90

L1 NAVIGATION PNT EXTRACTION — CURRENT LOCATION → NEW SERVICE LOCATION

Additive extension deliverable: each Position / Navigation / Timing behaviour wrapped by the AfsimL1 reusable service, showing its current in-controller location and accessor alongside the new injectable service member it is refactored into (see also the New Service Location column on Table 2).

PNT Pillar	Behavior	Current Location(s)	Current Accessor	New Service Location
Position	Read vehicle position	AP_L1_Control.cpp:L230, L369	_ahrs.get_location(_current_loc)	AHRS shim get_location(), fed by set_state_ne(n, e, ...)
Navigation (velocity)	Read ground velocity vector	AP_L1_Control.cpp:L236, L375, L507	_ahrs.groundspeed_vector()	AHRS shim groundspeed_vector(), fed by set_velocity_EN(veIE, veIN)
Navigation (attitude)	Read yaw (radians)	AP_L1_Control.cpp:L59, L61, L402, L536	_ahrs.get_yaw_rad()	AHRS shim get_yaw_rad(), fed by set_yaw_cd(yaw_cd)
Navigation (attitude)	Read yaw (centideg sensor)	AP_L1_Control.cpp:L70, L72, L503, L535	_ahrs.yaw_sensor	AHRS shim yaw_sensor, fed by set_yaw_cd(yaw_cd)
Navigation (attitude)	Read pitch (radians)	AP_L1_Control.cpp:L91	_ahrs.get_pitch_rad()	AHRS shim get_pitch_rad(), fed by set_pitch_rad(pitch_rad)
Navigation (airspeed)	Airspeed scaling factor	AP_L1_Control.cpp:L126, L159	_ahrs.get_EAS2TAS()	AHRS shim get_EAS2TAS() (injected or unit default)
Timing	Control-step clock	AP_L1_Control.cpp:L214	AP_HAL::micros()	set_update_dt(dt) injected timebase
Timing	Step delta + state store	AP_L1_Control.cpp:L215, L224	_last_update_waypoint_us	Injected-dt path (clamp preserved)
Timing	Loiter/heading clock	AP_L1_Control.cpp:L448	AP_HAL::millis()	Injected time for the loiter path

PNT Pillar	Behavior	Current Location(s)	Current Accessor	New Service Location
Navigation (math)	Bearing / NE distance	AP_L1_Control.cpp:L239, L382 / L260, L266, L274, L295, L391	Location::get_bearing_to / get_distance_NE	Preserved unchanged inside AP_L1_Control
Navigation (output)	Roll command	AP_L1_Control.h:L36	nav_roll_cd()	get_roll_deg() = nav_roll_cd()/100 → L1_GetRollDeg
Navigation (output)	Lateral acceleration	AP_L1_Control.h:L37	lateral_acceleration()	get_lat_accel() → L1_GetLatAccel